MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, MAGNITUDE, DIGITAL, KEY CODE JBK

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic digital magnitude comparator module.

1.2 Classification. This specification covers one class of modules.

Class 1 as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.11 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JBK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/42-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

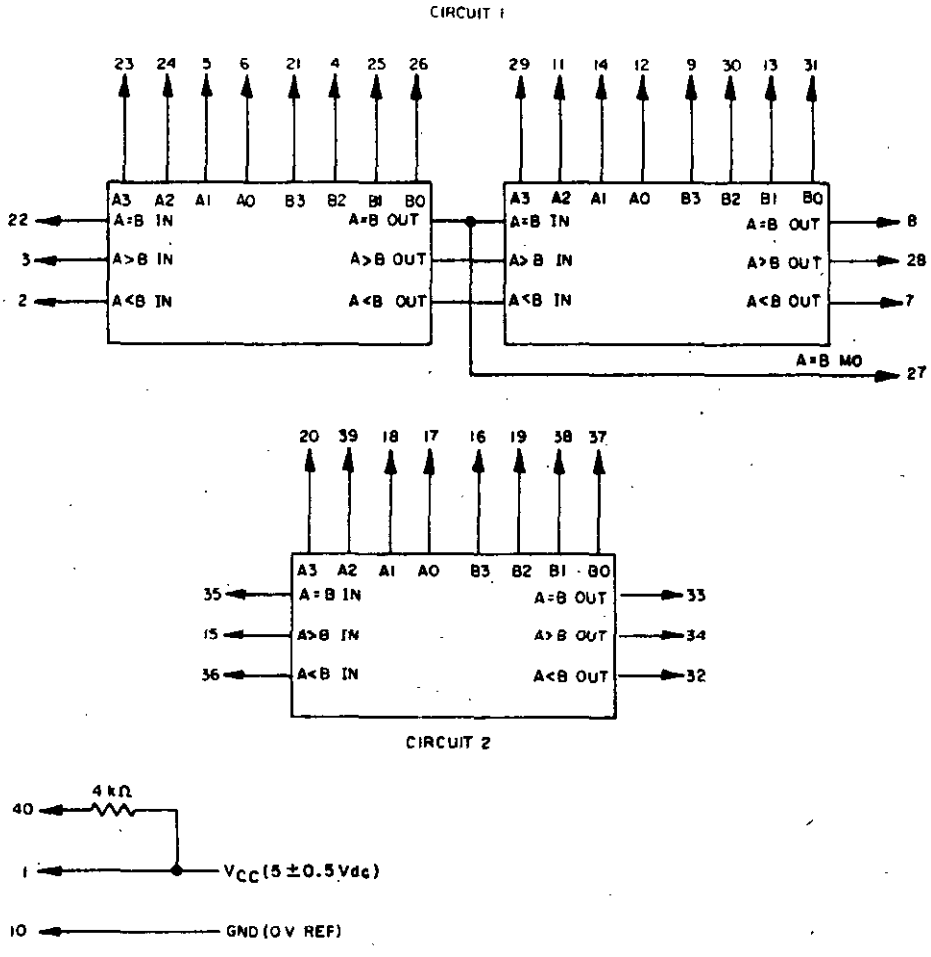
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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Circuit 1 - 8-bit magnitude comparator.
Circuit 2 - 4-bit magnitude comparator.
A0-A3, B0-B3 - Comparing inputs
A=B IN, A>B IN, A<B IN - Casading inputs.
A=B OUT, A>B OUT, A<B OUT - Outputs.
A=B MON - Monitor output.

FIGURE 1. Module function diagram, digital magnitude comparator.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A7	V_{OUT} (low-loaded)	III	S	0.01	0.6	0.01	0.6	0.01	0.6	0.01	0.6	V
B	Isolation	IV	S	10	∞	1	∞	10	∞	1	∞	MΩ
C1-C24	Static input current	V	S	0	300	0	300	0	300	0	300	µA
C25-C30	Static input current	V	S	0	100	0	100	0	100	0	100	µA
D1-D24	Input current, high level	VI	X	0	30	0	30	0	30	0	30	µA
D25-D30	Input current, high level	VI	X	0	10	0	10	0	10	0	10	µA
E	Pullup current	VI	X	1.30	1.45	1.30	1.45	1.30	1.50	1.30	1.50	mA
F1-F24	Input current, low level	VI	X	0.01	0.60	0.01	0.60	0.01	0.60	0.01	0.60	mA
F25-F30	Input current, low level	VI	X	0.01	0.20	0.01	0.20	0.01	0.20	0.1	0.20	mA
G1-G7	Output short circuit current	VI	X	3	15	3	15	3	15	3	15	mA
H1-H54	Functional compare	VII	X	← See table VII →								
J1-J7	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
K1-K7	Output voltage, low level	VII	X	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	V
L	Power supply current	VIII	S	5	24	5	24	5	24	5	24	mA
M1-M7	Transition time (t_{TLH}), C_L = 50 pF	IX	S	5	70	5	70	5	70	5	70	ns
M8-M14	Transition time (t_{TLH}), C_L = 150 pF	IX	Q	10	100	10	100	10	100	10	100	ns
N1-N7	Transition time (t_{THL}), C_L = 50 pF	IX	S	2	20	2	20	2	20	2	20	ns
N8-N14	Transition time (t_{THL}), C_L = 150 pF	IX	Q	3	30	3	30	3	30	3	30	ns
P1-P3, P40-P103	Propagation delay time (t_{PLH}), C_L = 50 pF	IX	S	10	300	10	300	10	320	10	320	ns
P4-P39	Propagation delay time (t_{PLH}), C_L = 50 pF	IX	S	10	150	10	150	10	170	10	170	ns
Q1-Q3	Propagation delay time (t_{PHL}), C_L = 50 pF	IX	S	10	200	10	200	10	220	10	220	ns
Q4-Q7	Propagation delay time (t_{PHL}), C_L = 50 pF	IX	S	10	100	10	100	10	120	10	120	ns

otnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q8-Q39	Propagation delay time (t_{PHL}), C_L = 50 pF	IX	S	10	150	10	150	10	170	10	170	ns
Q40-Q103	Propagation delay time (t_{PHL}), C_L = 50 pF	IX	S	10	300	10	300	10	320	10	320	ns
R1-R3, R40-R103	Propagation delay time (t_{PLH}), C_L = 150 pF	IX	Q	10	320	10	320	10	340	10	340	ns
R4-R39	Propagation delay time (t_{PLH}), C_L = 150 pF	IX	Q	10	170	10	170	10	190	10	190	ns
S1-S3	Propagation delay time (t_{PHL}), C_L = 150 pF	IX	Q	10	220	10	220	10	240	10	240	ns
S4-S7	Propagation delay time (t_{PHL}), C_L = 150 pF	IX	Q	10	120	10	120	10	140	10	140	ns
S8-S39	Propagation delay time (t_{PHL}), C_L = 150 pF	IX	Q	10	170	10	170	10	190	10	190	ns
S40-S103	Propagation delay time (t_{PHL}), C_L = 150 pF	IX	Q	10	320	10	320	10	340	10	340	ns
T	Power supply filter capacitance	X	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
U1-U7	Output crosstalk, high level	XI	Q	0	400	0	400	0	400	0	400	mV
V1-V7	Output crosstalk, low level	XI	Q	0	400	0	400	0	400	0	400	mV
W	Power supply transient amplitude	XII	Q	0	400	0	400	0	400	0	400	mV
X1-X6	Input capacitance	XIII	Q	2	15	2	15	2	15	2	15	pF
X7-X30	Input capacitance	XIII	Q	2	30	2	30	2	30	2	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever, is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through W, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/42-1 supersedes part number M28787/42. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	132 mW (outputs not loaded)
Supply voltage	5 Vdc ±10%
Input voltage	
Low level	-0.5 V minimum, 0.7 maximun
High level	2.0 V minimum, V_{CC} maximum

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Input current, low level		
(A and B inputs)		
(V_{IL} = 0.0 V)	-0.60 mA maximum	
(V_{IL} = 0.3 V)	-0.54 mA maximum	
(A = B, A > B, and A < B inputs)		
(V_{IL} = 0.0 V)	-0.20 mA maximum	
(V_{IL} = 0.3 V)	-0.18 mA maximum	
Input current, high level		
(A and B inputs)		
(V_{IH} = 2.4 V)	30 µA maximum	
(V_{IH} = 5.5 V)	300 µA maximum	
(A = B, A > B, and A < B inputs)		
(V_{IH} = 2.4 V)	10 µA maximum	
(V_{IH} = 5.5 V)	100 µA maximum	
Output voltage		
Low level (I_{OL} = 2.0 mA)	0.0 V minimum, 0.3 V maximum	
High level (I_{OH} = -100 µA)	2.4 V minimum, V_{CC} maximum	
Output drive current maximum capability (except A = B MON)		
Low Level (V_{OL} = 0.3 V)	2.0 mA	
High level (V_{OH} = 2.4 V)	-100 µA	
Output drive current maximum capability (A = B MON)		
Low level (V_{OL} = 0.3 V)	1.82 mA	
High level (V_{OH} = 2.4 V)	-90 µA	
Propagation delay times (circuit 1)	(C_L = 50 pF)	(C_L = 150 pF)
A < B IN to A < B OUT t_{PLH}	320 ns maximum	340 ns maximum
A = B IN to A = B OUT t_{PLH}	320 ns maximum	340 ns maximum
A > B IN to A > B OUT t_{PLH}	320 ns maximum	340 ns maximum
A IN > B IN t_{PLH} and t_{PHL}	320 ns maximum	340 ns maximum
A IN = B IN t_{PLH} and t_{PHL}	320 ns maximum	340 ns maximum
A IN < B IN t_{PLH} and t_{PHL}	320 ns maximum	340 ns maximum
A < B IN to A < B OUT t_{PHL}	220 ns maximum	240 ns maximum
A = B IN to A = B OUT t_{PHL}	220 ns maximum	240 ns maximum
A > B IN to A > B OUT t_{PHL}	220 ns maximum	240 ns maximum
Propagation delay times (circuit 2)		
A < B IN to A < B OUT t_{PLH}	170 ns maximum	190 ns maximum
A = B IN to A = B OUT t_{PLH}	170 ns maximum	190 ns maximum
A > B IN to A > B OUT t_{PLH}	170 ns maximum	190 ns maximum
A IN > B IN t_{PLH} and t_{PHL}	170 ns maximum	190 ns maximum
A IN = B IN t_{PLH} and t_{PHL}	170 ns maximum	190 ns maximum
A IN < B IN t_{PLH} and t_{PHL}	170 ns maximum	190 ns maximum
A < B IN to A < B OUT t_{PHL}	120 ns maximum	140 ns maximum
A = B IN to A = B OUT t_{PHL}	120 ns maximum	140 ns maximum
A > B IN to A > B OUT t_{PHL}	120 ns maximum	140 ns maximum

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6.5 Module function. This module consists of a 4-bit magnitude comparator and an 8-bit magnitude comparator. Each comparator has A = B, A > B, and A < B inputs and outputs. The 8-bit comparator also has the A = B output from the first stage brought out as an additional monitor of comparator results. The 4-bit comparator has four A and four B compare inputs; and the 8-bit comparator has eight A and eight B compare inputs. Both comparators perform magnitude comparison of straight binary or straight BCD (8421) codes. The 4 kΩ resistor is brought out as a logic circuit pullup resistor and used to allow pin 22 to be connected to the V_{CC}. This is required when the first stage of the 8-bit comparator is used as the least significant bit comparator.

6.5.1 Input to output relationship. The magnitude comparator module has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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TABLE II. Input to output relationship. 1/-3/

Comparing inputs												Cascading inputs			Outputs		
A11,B11	A10,B10	A9,B9	A8,B8	A7,B7	A6,B6	A5,B5	A4,B4	A3,B3	A2,B2	A1,B1	A0,B0	A>B	A<B	A=B	A>B	A<B	A=B
A11>B11	X	X	X									X	X	X	H	L	L
A11<B11	X	X	X									X	X	X	L	H	L
A11=B11	A10>B10	X	X									X	X	X	H	L	L
A11=B11	A10<B10	X	X									X	X	X	L	H	L
A11=B11	A10=B10	A9>B9	X									X	X	X	H	L	L
A11=B11	A10=B10	A9<B9	X									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8>B8									X	X	X	H	L	L
A11=B11	A10=B10	A9=B9	A8<B8									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									H	L	L	H	L	L
A11=B11	A10=B10	A9=B9	A8=B8									L	H	L	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									L	L	H	L	L	H
				A7>B7	X	X	X	X	X	X	X	X	X	X	H	L	L
				A7<B7	X	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6>B6	X	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6<B6	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5>B5	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5<B5	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4>B4	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4<B4	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3>B3	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3<B3	X	X	X	X	X	X	L	H	L

See footnotes at end of table.

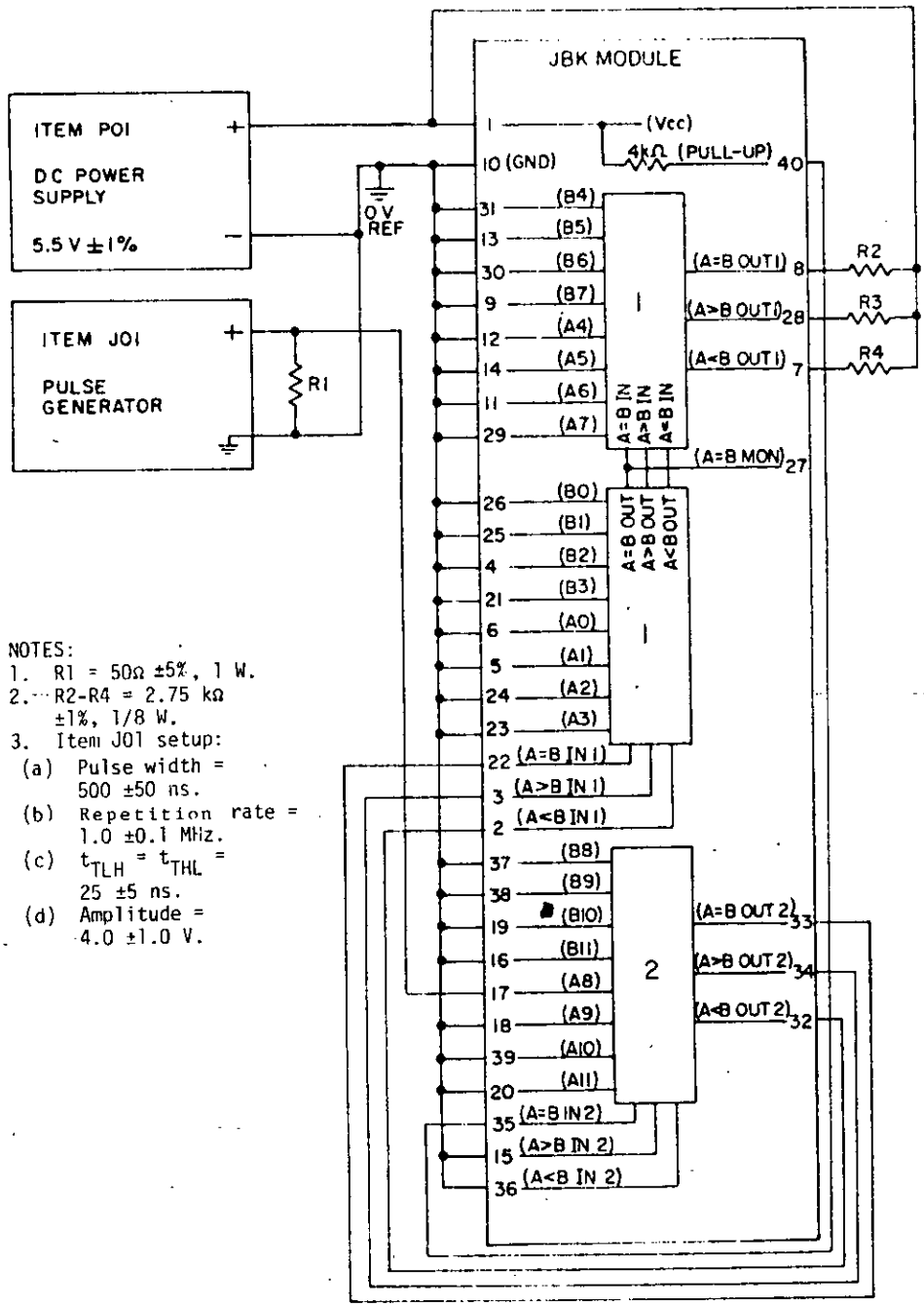
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TABLE II. Input to output relationship - Continued. 1/-3/

Comparing inputs												Cascading inputs			Outputs		
A11,B11	A10,B10	A9,B9	A8,B8	A7,B7	A6,B6	A5,B5	A4,B4	A3,B3	A2,B2	A1,B1	A0,B0	A>B	A<B	A=B	A>B	A<B	A=B
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2>B2	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2<B2	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1>B1	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1<B1	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0>B0	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0<B0	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	H	L	L	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	H	L	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	L	H	L	L	H

1/ X = Do not care term.
2/ H = High level voltage.
3/ L = Low level voltage.

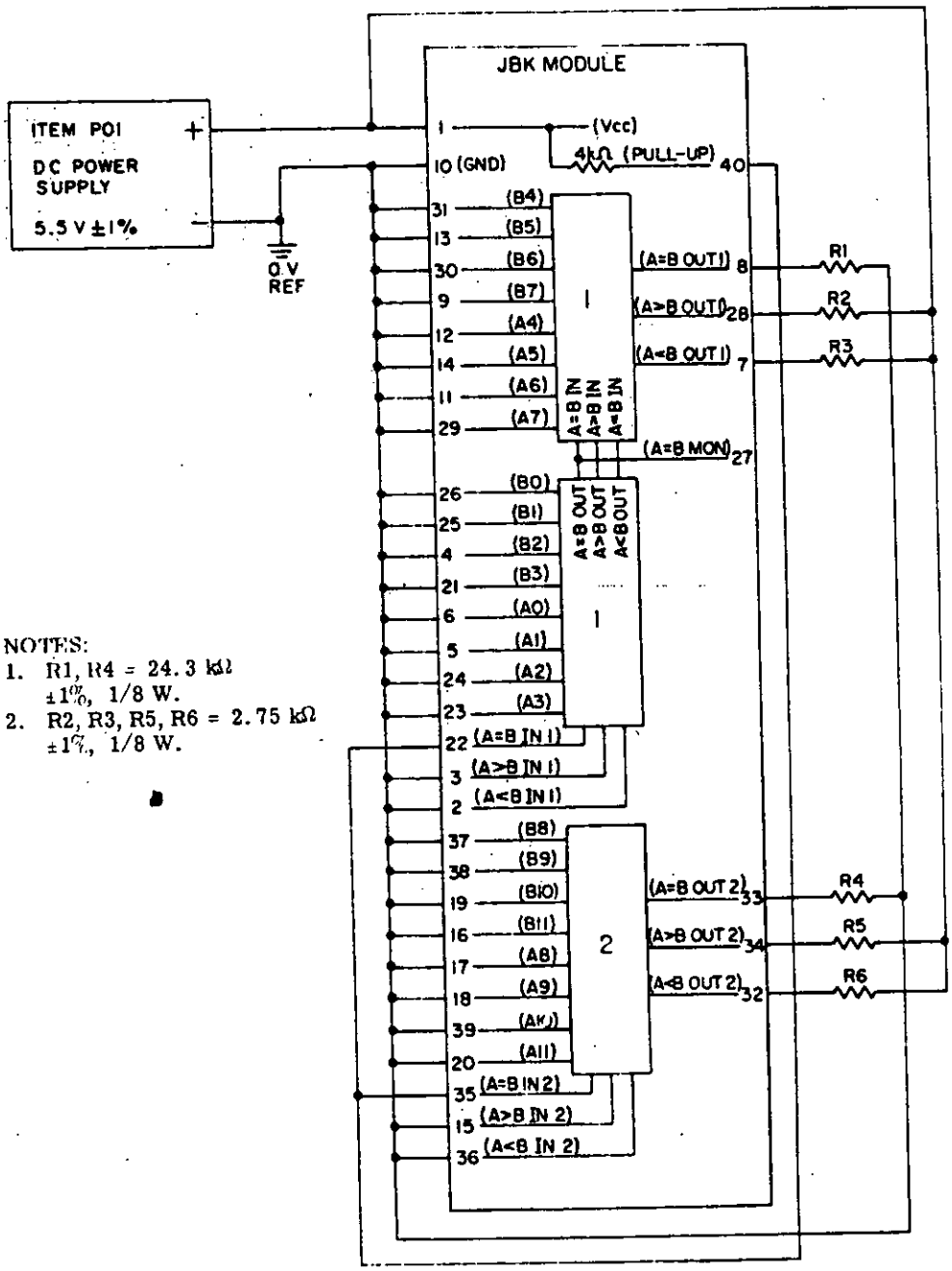
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2-R4 = 2.75 kΩ ±1%, 1/8 W.
3. Item J01 setup:
   (a) Pulse width = 500 ±50 ns.
   (b) Repetition rate = 1.0 ±0.1 MHz.
   (c) $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   (d) Amplitude = 4.0 ±1.0 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1, R4 = 24.3 kΩ ±1%, 1/8 W.
2. R2, R3, R5, R6 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 3. Transient thermal test circuit.

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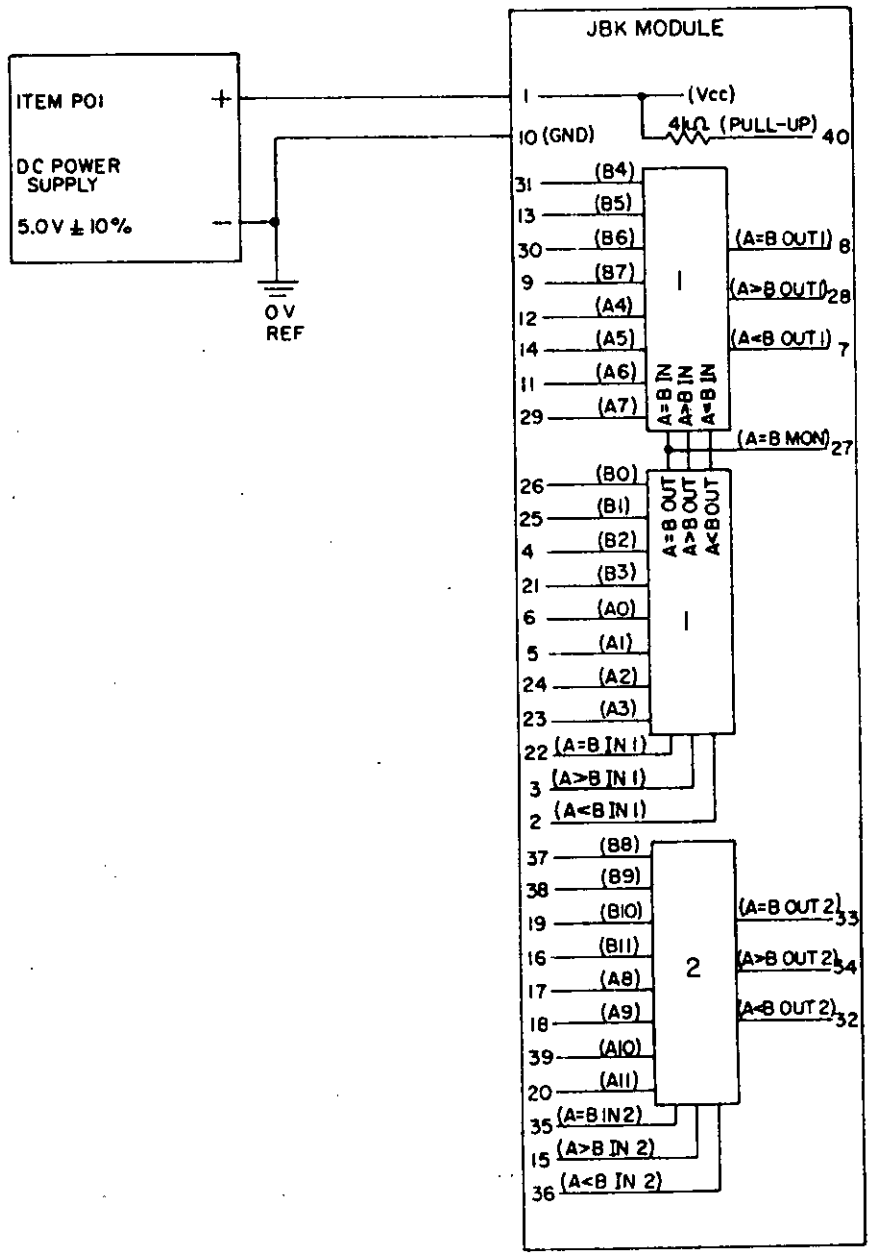


FIGURE 4. Vibration test circuit.

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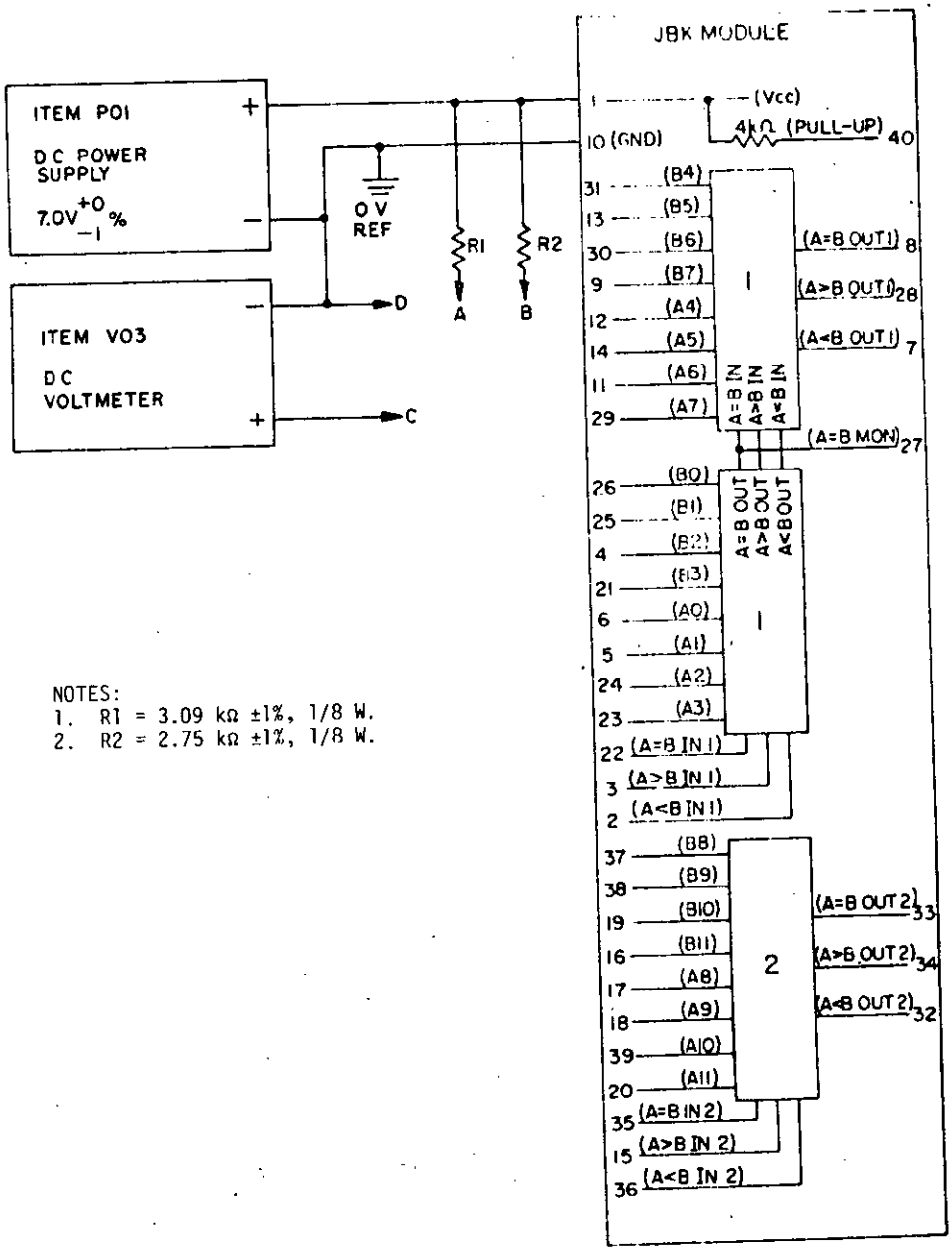
TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
A1	A>B OUT 1 V_{OUT} (low-loaded)	Open	28	28	29	0.01	0.6	V	5
A2	A<B OUT 1	Open	7	7	9				
A3	A=B OUT 1	Open	8	8	9				
A4	A=B MON 1	27	Open	27	21				
A5	A>B OUT 2	Open	34	34	20				
A6	A<B OUT 2	Open	32	32	16				
A7	A=B OUT 2	Open	33	33	16				

TABLE IV. Isolation test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B	Isolation	Frame	10	∞	MΩ	6

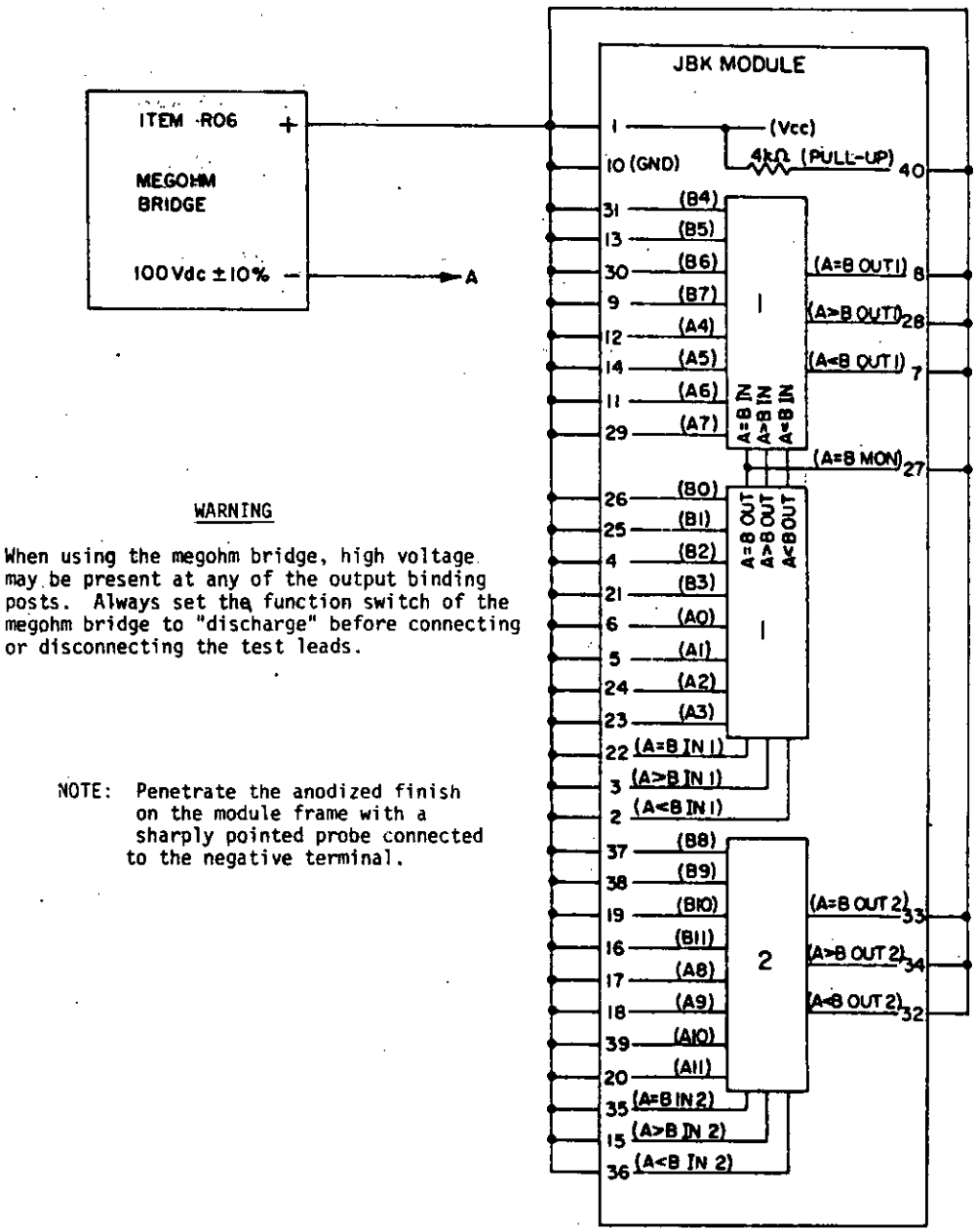
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NOTES:
1. R1 = 3.09 kΩ ±1%, 1/8 W.
2. R2 = 2.75 kΩ ±1%, 1/8 W.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 6. Isolation test circuit for table IV.

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TABLE V. Static input current test.

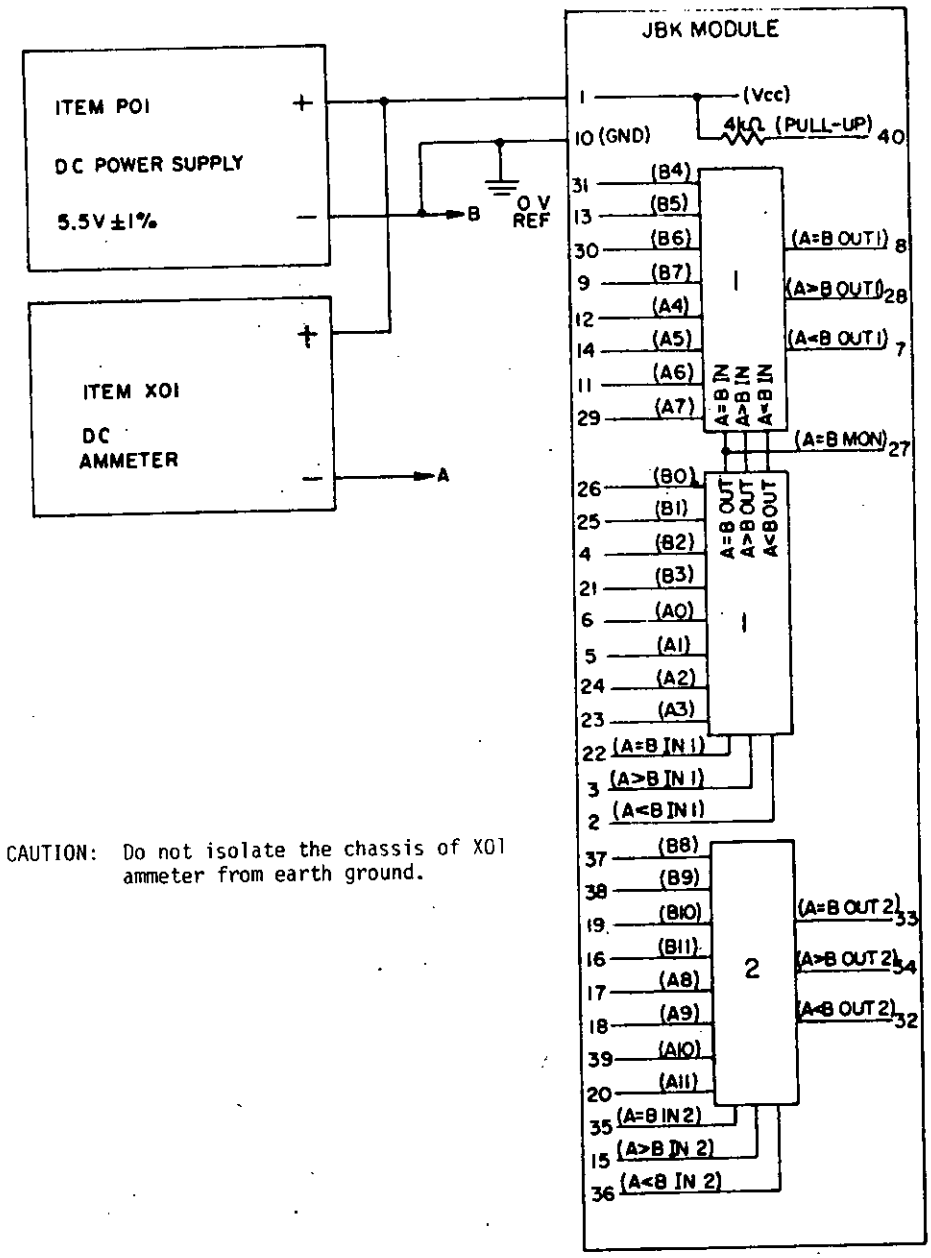
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A0 Static input current	6	Open	0	300	µA	7
C2	A1	5					
C3	A2	24					
C4	A3	23					
C5	A4	12					
C6	A5	14					
C7	A6	11					
C8	A7	29					
C9	A8	17					
C10	A9	18					
C11	A10	39					
C12	A11	20					
C13	B0	26					
C14	B1	25					
C15	B2	4					
C16	B3	21					
C17	B4	31					
C18	B5	13					
C19	B6	30					
C20	B7	9					
C21	B8	37					
C22	B9	38					
C23	B10	19					
C24	B11	16					
C25	A=B IN	22	26		100		

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TABLE V. Static input current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C26	A>B IN 1 Static input current	3	26	0	100	µA	7
C27	A<B IN 1	2	26	↓	↓	↓	↓
C28	A=B IN 2	35	37				
C29	A>B IN 2	15	37				
C30	A<B IN 2	36	37				

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CAUTION: Do not isolate the chassis of X01 ammeter from earth ground.

FIGURE 7. Static input current test circuit for table V.

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TABLE VI.. Input current (high-level and low-level), pullup current, and output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	A0 Input current, high level	6	Open	Open	0	30	µA	8
D2	A1	5						
D3	A2	24						
D4	A3	23						
D5	A4	12						
D6	A5	14						
D7	A6	11						
D8	A7	29						
D9	A8	17						
D10	A9	18						
D11	A10	39						
D12	A11	20						
D13	B0	26						
D14	B1	25						
D15	B2	4						
D16	B3	21						
D17	B4	31						
D18	B5	13						
D19	B6	30						
D20	B7	9						
D21	B8	37						
D22	B9	38						
D23	B10	19						
D24	B11	16						

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TABLE VI. Input current (high-level and low-level), pullup current, and output short circuit current test - Continued.

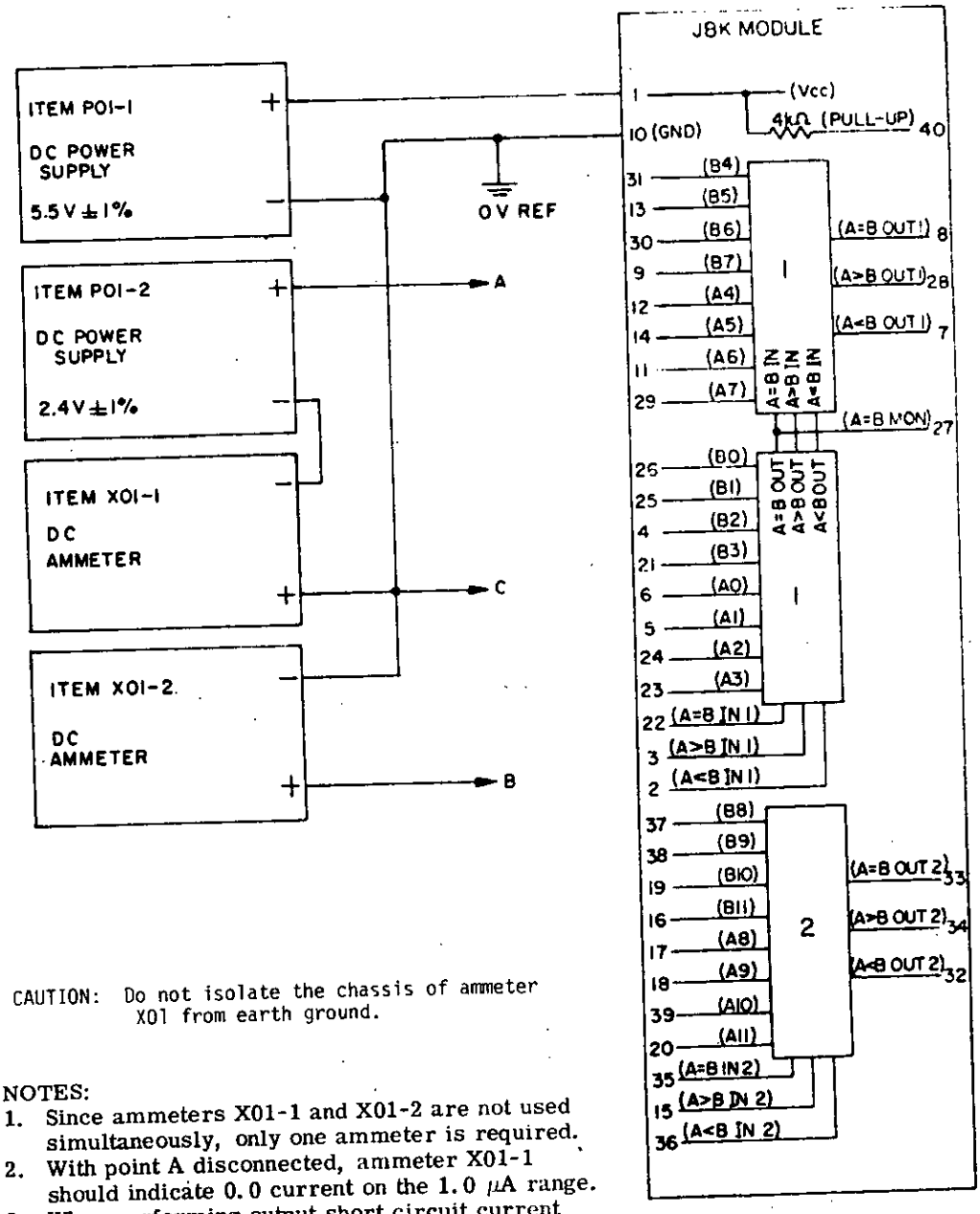
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D25	A=B IN 1 Input current, high level	22	Open	26	0	10	μA	8
D26	A=B IN 2	35		37				
D27	A>B IN 1	3		26				
D28	A<B IN 1	2		26				
D29	A>B IN 2	15		37				
D30	A<B IN 2	36		37				
E	Pullup current	Open	40	Open	1.30	1.45	mA	
F1	A0 Input current, low level		6	26	0.01	0.60		
F2	A1		5	25				
F3	A2		24	4				
F4	A3		23	21				
F5	A4		12	31				
F6	A5		14	13				
F7	A6		11	30				
F8	A7		29	9				
F9	A8		17	37				
F10	A9		18	38				
F11	A10		39	19				
F12	A11		20	16				
F13	B0		26	6				
F14	B1		25	5				
F15	B2		4	24				
F16	B3		21	23				

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TABLE VI. Input current (high-level and low-level), pullup current, and output short circuit current test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F17	B4	Input current,low level	Open	31	12	0.01	0.60	mA	8
F18	B5			13	14				
F19	B6			30	11				
F20	B7			9	29				
F21	B8			37	17				
F22	B9			38	18				
F23	B10			19	39				
F24	B11			16	20				
F25	A=B IN 1			22	Open		0.20		
F26	A=B IN 2			35					
F27	A>B IN 1			3					
F28	A<B IN 1			2					
F29	A>B IN 2			15					
F30	A<B IN 2			36					
G1	A=B OUT 1	Output short circuit current	22	8	2, 3	3	15		
G2	A>B OUT 1		3	28	2,22				
G3	A<B OUT 1		2	7	3,22				
G4	A=B OUT 2		35	33	15,36				
G5	A>B OUT 2		15	34	35,36				
G6	A<B OUT 2		36	32	15,35				
G7	A=B MON		22	27	2, 3				

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CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since ammeters X01-1 and X01-2 are not used simultaneously, only one ammeter is required.
2. With point A disconnected, ammeter X01-1 should indicate 0.0 current on the 1.0 μA range.
3. When performing output short circuit current test, only one output shall be shorted at a time, and the measurement taken within 10 seconds.

FIGURE 8. Input current (high-level and low-level), pullup current, and output short circuit current test circuit for table VI.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H1	Functional compare (circuit 2)	35	20	16	2/	15,36	L	H	L	--	--	--	9
H2			16	20			L	L	H				
H3			20,16	Open			H	L	L				
H4			Open	20,16			H	L	L				
H5			39	19		15,16, 20,36	L	H	L				
H6			19	39			L	L	H				
H7			19,39	Open			H	L	L				
H8			Open	19,39			H	L	L				
H9			18	38		15,16, 19,20, 36,39	L	H	L				
H10			38	18			L	L	H				
H11			18,38	Open			H	L	L				
H12			Open	18,38			H	L	L				
H13			17	37		15,16, 18,19, 20,36, 38,39	L	H	L				
H14			37	17		15,16, 18,19, 20,36, 38,39	L	L	H				

See footnotes at end of table.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H15	Functional compare (circuit 2)	35	17,37	Open	2/	15,16, 18,19, 20,36, 38,39	H	L	L	--	--	--	9
H16		35	Open	17,37		15,16, 18,19, 20,36, 38,39	H	L	L				
H17		Open	35	15,36		16,17, 18,19, 20,37, 38,39	H	L	L				
H18		Open	15	35,36		16,17, 18,19, 20,37, 38,39	L	H	L				
H19		Open	36	15,35		16,17, 18,19, 20,37, 38,39	L	L	H				
H20	Functional compare (circuit 1)	22	29	9		2,3	L	H	L				
H21			9	29			L	L	H				
H22			9,29	Open			H	L	L				
H23			Open	9,29			H	L	L				
H24			11	30		2,3, 9,29	L	H	L				
H25			30	11			L	L	H				
H26			11,30	Open			H	L	L				
H27			Open	11,30			H	L	L				

See footnotes at end of table.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H28	Functional compare (circuit 1)	22	14	13	2/	2,3, 9,11, 29,30	L	H	L	--	--	--	9
H29			13	14			L	L	H				
H30			13,14	Open			H	L	L				
H31			Open	13,14			H	L	L				
H32			12	31		2,3,9, 11,13, 14,29, 30	L	H	L				
H33			31	12			L	L	H				
H34			12,31	Open			H	L	L				
H35			Open	12,31			H	L	L				
H36			23	21		2,3,9, 11,12, 13,14, 29,30, 31	L	H	L				

See footnotes at end of table.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H37	Functional compare (circuit 1)	22	21	23	2/	2,3,9, 11,12, 13,14, 29,30, 31	L	L	H	--	--	--	9
H38			21,23	Open		2,3,9, 11,12, 13,14, 29,30, 31	H	L	L				
H39			Open	21,23		2,3,9, 11,12, 13,14, 29,30, 31	H	L	L				
H40			24	4		2,3,9, 11,12, 13,14, 21,23, 29,30 31	L	H	L				
H41			4	24		2,3,9, 11,12, 13,14, 21,23, 29,30, 31	L	L	H				
H42			4,24	Open		2,3,9, 11,12, 13,14, 21,23, 29,30, 31	H	L	L				

See footnotes at end of table.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H43	Functional compare (circuit 1)	22	Open	4,24	2/	2,3,9, 11,12, 13,14, 21,23, 29,30, 31	H	L	L	--	--	--	9
H44			5	25		2,3,4, 9,11, 12,13, 14,21, 23,24, 29,30, 31	L	H	L				
H45			25	5			L	L	H				
H46			5,25	Open			H	L	L				
H47			Open	5,25			H	L	L				

See footnotes at end of table.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H48	Functional compare (circuit 1)	22	6	26	2/	2,3,4, 5,9, 11,12, 13,14, 21,23, 24,25, 29,30, 31	L	H	L	--	--	--	9
H49			26	6			L	L	H				
H50			6,26	Open			H	L	L				
H51			Open	6,26			H	L	L				
H52		Open	22	2,3		4,5,6, 9,11, 12,13, 14,21, 23,24, 25,26, 29,30, 31	H	L	L				

See footnotes at end of table.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
H53	Functional compare (circuit 1)	Open	3	2,22	2/	4,5,6, 9,11, 12,13, 14,21, 23,24, 25,26, 29,30, 31	L	H	L	--	--	--	9
H54	Functional compare (circuit 1)		2	3,22	2/	4,5,6, 9,11, 12,13, 14,21, 23,24, 25,26, 29,30, 31	L	L	H	--	--	--	
J1	A=B OUT 2 Output voltage, high level		35	Open	33	15,16, 17,18, 19,20, 36,37, 38,39	H	L	L	2.4	4.5	V	
J2	A>B OUT 2		15		34	35,36	L	H	L				
J3	A<B OUT 2		36		32	15,16, 17,18, 19,20, 35,37, 38,39	L	L	H				
J4	A=B OUT 1		22		8	2,3,4, 5,6,9, 11,12, 13,14, 21,23, 24,25, 26,29, 30,31	H	L	L				

See footnotes at end of table.

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TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
J5	A=B MON Output voltage, high level	Open	22	Open	27	2,3,4, 5,6,9, 11,12, 13,14, 21,23, 24,25, 26,29, 30,31	H	L	L	2.4	4.5	V	9
J6	A>B OUT 1 Output voltage, high level	Open	3	Open	28	2,22	L	H	L	2.4	4.5		
J7	A<B OUT 1 Output voltage, high level	Open	2	Open	7	3,4,5, 6,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	L	L	H	2.4	4.5		
K1	A=B OUT 2 Output voltage, low level	36	Open	35	33	15,16, 17,18, 19,20, 37,38, 39	L	L	H	0.0	0.3		
K2	A>B OUT 2 Output voltage, low level	35	Open	15	34	36	H	L	L	0.0	0.3		
K3	A<B OUT 2 Output voltage, low level	15	Open	36	32	16,17, 18,19, 20,35, 37,38, 39	L	H	L	0.0	0.3		

See footnotes at end of table.

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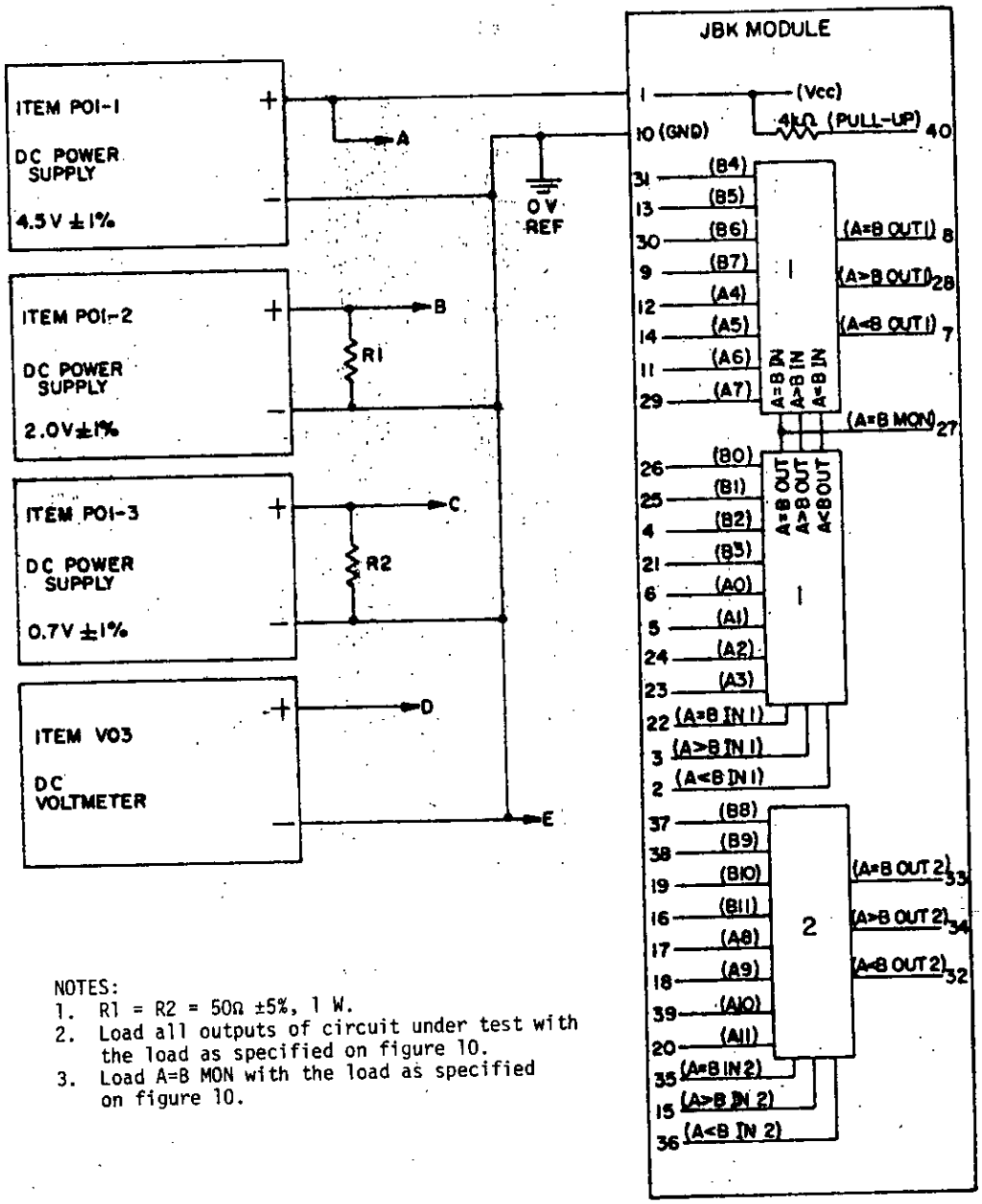
TABLE VII. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect					Outputs 1/			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	A=B	A>B	A<B	Min	Max		
K4	A=B OUT 1 Output voltage, low level	2	Open	22	8	3,4,5, 6,9, 11,12, 13,14, 21,23, 24,25, 26,29, 30,31	L	L	H	0.0	0.3	V	9
K5	A=B MON	2		22	27	3,4,5, 6,9, 11,12, 13,14, 21,23, 24,25, 26,29, 30,31	L	L	H				
K6	A>B OUT 1	22		2	28	3,4,5, 6,9, 11,12, 13,14, 21,23, 24,25, 26,29, 30,31	H	L	L				
K7	A<B OUT 1	3		2	7	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 26,29, 30,31	L	H	L				

1/ Output logic states: L = low level voltage = 0.0 to 0.3 V.
H = high level voltage = 2.4 to 4.5 V.

2/ For test conditions H1-H54, verification of the output states specified in footnote 1 is obtained by connecting voltmeter V03 (point D) to each output pin after input conditions are accomplished.

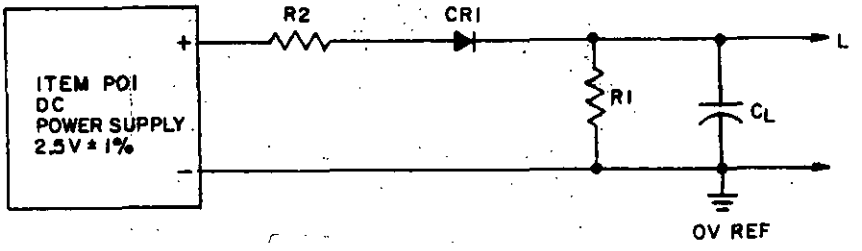
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load all outputs of circuit under test with the load as specified on figure 10.
3. Load A=B MON with the load as specified on figure 10.

FIGURE 9. Functional compare and output voltage (high-level and low-level) test circuit for table VII.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. C_L = 50 ±5 pF (including probe and parasitics) for test conditions: M1-M7, N1-N7, P, Q, U, and V.
3. C_L = 150 ±5 pF (including probe and parasitics) for test conditions: M8-M14, N8-N14, R, and S.
4. R1 = 24.3 kΩ ±1%, 1/8 W (all outputs except A = B MON).
5. For A = B MON load, R1 = 26.7 kΩ ±1%, 1/8 W.
6. R2 = Select value for a current flow of 2.0 mA ±5% out of point L when point L is held at 0.3 V, (all outputs except A = B MON).
7. For A = B MON load, R2 = Select value for a current flow of 1.82 mA ±5% out of point L when point L is held at 0.3 V.
8. The capacitor is not required for test conditions H, J, and K.

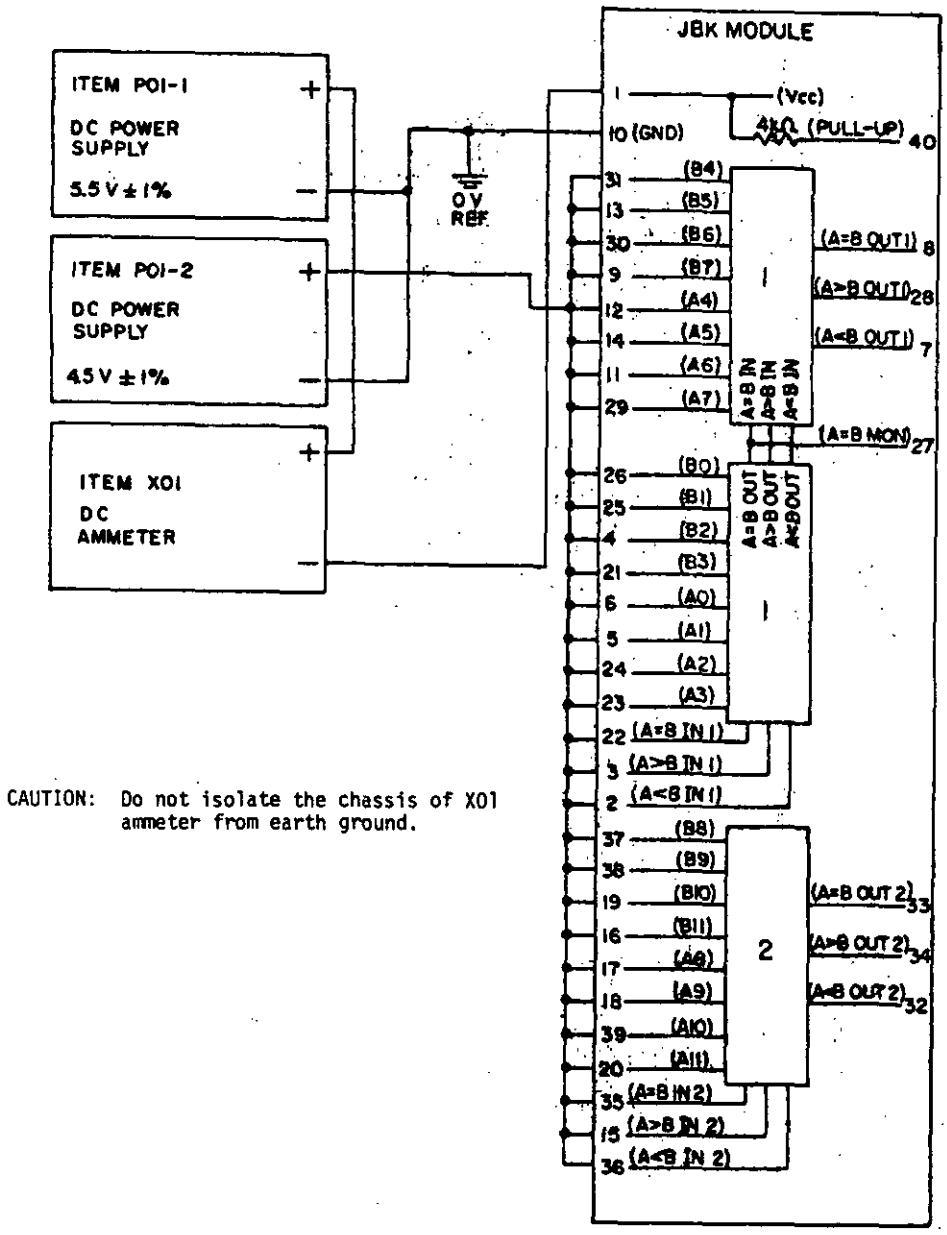
FIGURE 10. Logic load circuit.

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TABLE VIII. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	5	24	mA	11

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CAUTION: Do not isolate the chassis of X01 ammeter from earth ground.

FIGURE 11. *Power supply current test circuit for table VIII.*

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	A>B OUT 1 Transition time (t_{TLH}) C_L = 50 pF	Open	3	28	2,4,5,6,9,11,12, 13,14,21,22,23,24 25,26,29,30,31	5	70	ns	12, 13
M2	A<B OUT 1		2	7	3,4,5,6,9,11,12, 13,14,21,22,23,24 25,26,29,30,31				
M3	A=B OUT 1		22	8	2,3,4,5,6,9,11,12 13,14,21,23,24,25 26,29,30,31				
M4	A=B MON		22	27	2,3,4,5,6,9,11,12 13,14,21,23,24,25 26,29,30,31				
M5	A>B OUT 2		15	34	16,17,18,19,20,35 36,37,38,39				
M6	A<B OUT 2		36	32	15,16,17,18,19,20 35,37,38,39				
M7	A=B OUT 2		35	33	15,16,17,18,19,20, 36,37,38,39				
M8	A>B OUT 1 Transition time (t_{TLH}), C_L = 150 pF		3	28	2,4,5,6,9,11,12, 13,14,21,22,23,24 25,26,29,30,31	10	100		
M9	A<B OUT 1		2	7	3,4,5,6,9,11,12, 13,14,21,22,23,24 25,26,29,30,31				
M10	A=B OUT 1		22	8	2,3,4,5,6,9,11,12 13,14,21,23,24,25 26,29,30,31				
M11	A=B MON		22	27	2,3,4,5,6,9,11,12 13,14,21,23,24,25 26,29,30,31				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M12	A>B OUT 2 Transition time (t_{TLH}), C_L = 150 pF	Open	15	34	16,17,18,19,20, 35,36,37,38,39	10	100	ns	12, 13
M13	A<B OUT 2 Transition time (t_{TLH}), C_L = 150 pF		36	32	15,16,17,18,19, 20,35,37,38,39	10	100		
M14	A=B OUT 2 Transition time (t_{TLH}), C_L = 150 pF		35	33	15,16,17,18,19, 20,36,37,38,39	10	100		
N1	A>B OUT 1 Transition time (t_{THL}), C_L = 50 pF		3	28	2,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31	2	20		
N2	A<B OUT 1		2	7	3,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31				
N3	A=B OUT 1		22	8	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30,31				
N4	A=B MON		22	27	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30,31				
N5	A>B OUT 2		15	34	16,17,18,19,20, 35,36,37,38,39				
N6	A<B OUT 2		36	32	15,16,17,18,19, 20,35,37,38,39				
N7	A=B OUT 2		35	33	15,16,17,18,19, 20,36,37,38,39				
N8	A>B OUT 1 Transition time (t_{THL}), C_L = 150 pF		3	28	2,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31	3	30		
N9	A<B OUT 1 Transition time (t_{THL}), C_L = 150 pF		2	7	3,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31	3	30		

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N10	A=B OUT 1 Transition time (t_{THL}), C_L = 150 pF	Open	22	8	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30,31	3	30	ns	12, 13
N11	A=B MON		22	27	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30,31				
N12	A>B OUT 2		15	34	16,17,18,19,20, 35,36,37,38,39				
N13	A<B OUT 2		36	32	15,16,17,18,19, 20,35,37,38,39				
N14	A=B OUT 2		35	33	15,16,17,18,19, 20,36,37,38,39				
P1	A<B IN 1 to A<B OUT 1 Propagation delay time (t_{PLH}), C_L = 50 pF		2	7	3,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31	10	300		
P2	A=B IN 1 to A=B OUT 1		22	8	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30,31		300		
P3	A>B IN 1 to A>B OUT 1		3	28	2,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31		300		
P4	A=B IN 1 to A=B MON		22	27	2,3,4,5,6,9,11, 12,13,14,21,23,24 25,26,29,30,31		150		
P5	A<B IN 2 to A<B OUT 2		36	32	15,16,17,18,19, 20,35,37,38,39		150		
P6	A=B IN 2 to A=B OUT 2		35	33	15,16,17,18,19, 20,36,37,38,39		150		

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P7	A>B IN 2 Propagation delay time to A>B OUT 2 (t_{PLH}), C_L = 50 pF	Open	15	34	16,17,18,19,20, 35,36,37,38,39	10	150	ns	12, 13
P8	A11>B11	35	20		15,16,36				
P9	A10>B10		39		15,16,19,20,36				
P10	A9>B9		18		15,16,19,20,36, 38, 39				
P11	A8>B8		17		15,16,18,19,20, 36,37,38,39				
P12	A11=B11		20	33	15,16,36				
P13	A10=B10		39		15,16,19,20,36				
P14	A9=B9		18		15,16,19,20,36, 38,39				
P15	A8=B8		17		15,16,18,19,20, 36,37,38,39				
P16	A11<B11	16,35	20	32	15,36				
P17	A10<B10	16,19, 20,35	39						
P18	A9<B9	16,19, 20,35, 38,39	18						
P19	A8<B8	16,18, 19,20, 35,37, 38,39	17						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P20	A11=B11 Propagation delay time (t_{PLH}), C_L = 50 pF	16,35	20	33	15,36	10	150	ns	12, 13
P21	A10=B10	16,19, 20,35	39						
P22	A9=B9	16,19, 20,35, 38,39	18						
P23	A8=B8	16,18, 19,20, 35,37, 38,39	17						
P24	A11>B11 Propagation delay time (t_{PLH}), pulse on B, 5 V on A, C_L = 50 pF	20,35	16	34					
P25	A10>B10	16,20, 39,35	19						
P26	A9>B9	16,18, 19,20, 35,39	38						
P27	A8>B8	16,17, 18,19, 20,35, 38,39	37						
P28	A11=B11	20,35	16	33					
P29	A10=B10	16,20, 35,39	19	33					
P30	A9=B9	16,18, 19,20, 35,39	38	33					

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P31	A8=B8 Propagation delay time (t_{PLH}), C_L = 50 pF	16,17, 18,19, 20,35, 38,39	37	33	15,36	10	150	ns	12, 13
P32	A11<B11	35	16	32	15,20,36				
P33	A10<B10		19		15,16,20,36,39				
P34	A9<B9		38		15,16,18,19,20, 36,39				
P35	A8<B8		37		15,16,17,18,19, 20,36,38,39				
P36	A11=B11		16	33	15,20,36				
P37	A10=B10		19		15,16,20,36,39				
P38	A9=B9		38		15,16,18,19,20, 36,39				
P39	A8=B8		37		15,16,17,18,19, 20,36,38,39				
P40	A7>B7	22	29	28	2,3,9		300		
P41	A6>B6		11		2,3,9,29,30				
P42	A5>B5		14		2,3,9,11,13,29,30				
P43	A4>B4		12		2,3,9,11,13,14, 29,30,31				
P44	A3>B3		23		2,3,9,11,12,13, 14,21,29,30,31				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P45	A2>B2 Propagation delay time (t_{PLH}), C_L = 50 pF	22	24	28	2,3,4,9,11,12, 13,14,21,23,29, 30,31	10	300	ns	12, 13
P46	A1>B1		5	28	2,3,4,9,11,12,13, 14,21,23,24,25, 29,30,31				
P47	A0>B0		6	28	2,3,4,5,9,11,12, 13,14,21,23,24, 25,26,29,30,31				
P48	A7=B7		29	8	2,3,9				
P49	A6=B6		11		2,3,9,29,30				
P50	A5=B5		14		2,3,9,11,13,29,30				
P51	A4=B4		12		2,3,9,11,13,14, 29,30,31				
P52	A3=B3		23		2,3,9,11,12,13, 14,21,29,30,31				
P53	A2=B2		24		2,3,4,9,11,12,13, 14,21,23,29,30,31				
P54	A1=B1		5		2,3,4,9,11,12,13, 14,21,23,24,25, 29,30,31				
P55	A0=B0		6		2,3,4,5,9,11,12, 13,14,21,23,24, 25,26,29,30,31				
P56	A7< B7	9,22	29	7	2,3				
P57	A6< B6	9,22, 29,30	11	7	2,3				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
P58	A5<B5 Propagation delay time (t_{PLH}), C_L = 50 pF	9,11, 13,22, 29,30	14	7	2,3	10	300	ns	12, 13
P59	A4<B4	9,11, 13,14, 22,29, 30,31	12						
P60	A3<B3	9,11, 12,13, 14,21, 22,29, 30,31	23						
P61	A2<B2	4,9, 11,12, 13,14, 21,22, 29,30, 31,23	24						
P62	A1<B1	4,9, 11,12, 13,14, 21,22, 23,24, 25,29, 30,31	5						
P63	A0<B0	4,5,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	6						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P64	A7=B7 Propagation delay time (t_{PLH}), C_L = 50 pF	9,22	29	8	2,3	10	300	ns	12, 13
P65	A6=B6	9,22, 29,30	11						
P66	A5=B5	9,11, 13,22, 29,30	14						
P67	A4=B4	9,11, 13,14, 22,29, 30,31	12						
P68	A3=B3	9,11, 12,13, 14,21, 22,29, 30,31	23						
P69	A2=B2	4,9, 11,12, 13,14, 21,22, 23,29, 30,31	24						
P70	A1=B1	4,9, 11,12, 13,14, 21,22, 23,24, 25,29, 30,31	5						
P71	A0=B0	4,5,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	6						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P72	A7>B7 Propagation delay time (t_{PLH}), C_L = 50 pF	22,29	9	28	2,3	10	300	ns	12, 13
P73	A6>B6	9,11, 22,29	30						
P74	A5>B5	9,11, 14,22, 29,30	13						
P75	A4>B4	9,11, 12,13, 14,22, 29,30	31						
P76	A3>B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
P77	A2>B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
P78	A1>B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						
P79	A0>B0	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	26						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P80	A7=B7 Propagation delay time (t_{PLH}), C_L = 50 pF	22,29	9	8	2,3	10	300	ns	12, 13
P81	A6=B6	9,11, 22,29	30						
P82	A5=B5	9,11, 14,22, 29,30	13						
P83	A4=B4	9,11, 12,13, 14,22, 29,30	31						
P84	A3=B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
P85	A2=B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
P86	A1=B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						
P87	A0=B0	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	26						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P88	A7<B7 Propagation delay time (t_{PLH}), C_L = 50 pF	22	9	7	2,3,29	10	300	ns	12, 13
P89	A6<B6		30		2,3,9,11,29				
P90	A5<B5		13		2,3,9,11,14,29,30				
P91	A4<B4		31		2,3,9,11,12,13, 14,29,30				
P92	A3<B3		21		2,3,9,11,12,13, 14,23,29,30,31				
P93	A2<B2		4		2,3,9,11,12,13, 14,21,23,24,29, 30,31				
P94	A1<B1		25		2,3,4,5,9,11,12, 13,14,21,23,24, 29,30,31				
P95	A0<B0		26		2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,29,30,31				
P96	A7=B7		9	8	2,3,29				
P97	A6=B6		30		2,3,9,11,29				
P98	A5=B5		13		2,3,9,11,14,29,30				
P99	A4=B4		31		2,3,9,11,12,13, 14,29,30				
P100	A3=B3		21		2,3,9,11,12,13, 14,23,29,30,31				
P101	A2=B2		4		2,3,9,11,12,13, 14,21,23,24,29, 30,31				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P102	A1=B1 Propagation delay time (t_{PLH}), C_L = 50 pF	22	25	8	2,3,4,5,9,11,12, 13,14,21,23,24, 29,30,31	10	300	ns	12, 13
P103	A0=B0 Propagation delay time (t_{PLH}), C_L = 50 pF	22	26	8	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,29,30,31		300		
Q1	A<B IN 1 to A<B OUT 1 Propagation delay time (t_{PHL}), C_L = 50 pF	Open	2	7	3,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31		200		
Q2	A=B IN 1 to A=B OUT 1		22	8	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30,31		200		
Q3	A>B IN 1 to A>B OUT 1		3	28	2,4,5,6,9,11,12, 13,14,21,22,23, 24,25,26,29,30,31		200		
Q4	A=B IN 1 to A=B MON		22	27	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30,31		100		
Q5	A<B IN 2 to A<B OUT 2		36	32	15,16,17,18,19, 20,35,37,38,39				
Q6	A=B IN 2 to A=B OUT 2		35	33	15,16,17,18,19, 20,36,37,38,39				
Q7	A>B IN 2 to A>B OUT 2		15	34	16,17,18,19,20, 35,36,37,38,39				
Q8	A11>B11	35	20	34	15,16,36		150		

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q9	A10>B10 Propagation delay time (t_{PHL}), C_L = 50 pF	35	39	34	15,16,19,20,36	10	150	ns	12, 13
Q10	A9>B9		18	34	15,16,19,20,36, 38,39				
Q11	A8>B8		17	34	15,16,18,19,20, 36,37,38,39				
Q12	A11=B11		20	33	15,16,36				
Q13	A10=B10		39		15,16,19,20,36				
Q14	A9=B9		18		15,16,19,20,36, 38,39				
Q15	A8=B8		17		15,16,18,19,20, 36,37,38,39				
Q16	A11<B11	16,35	20	32	15,36				
Q17	A10<B10	16,19, 20,35	39						
Q18	A9<B9	16,19, 20,35, 38,39	18						
Q19	A8<B8	16,18, 19,20, 35,37, 38,39	17						
Q20	A11=B11	16,35	20	33					

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q21	A10=B10 Propagation delay time (t_{PHL}), C_L = 50 pF	16,19, 20,35	39	33	15,36	10	150	ns	12, 13
Q22	A9=B9	16,19, 20,35, 38,39	18	33					
Q23	A8=B8	16,18, 19,20, 35,37, 38,39	17	33					
Q24	A11>B1	20,35	16	34					
Q25	A10>B10	16,20, 39,35	19						
Q26	A9>B9	16,18, 19,20, 35,39	38						
Q27	A8>B8	16,17, 18,19, 20,35, 38,39	37						
Q28	A11=B11	20,35	16	33					
Q29	A10=B10	16,20, 35,39	19						
Q30	A9=B9	16,18, 19,20, 35,39	38						
Q31	A8=B8	16,17, 18,19, 20,35, 38,39	37						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q32	A11<B11 Propagation delay time (t_{PHL}), C_L = 50 pF	35	16	32	15,20,36	10	150	ns	12, 13
Q33	A10<B10		19		15,16,20,36,39				
Q34	A9<B9		38		15,16,18,19,20, 36,39				
Q35	A8<B8		37		15,16,17,18,19, 20,36,38,39				
Q36	A11=B11		16	33	15,20,36				
Q37	A10=B10		19		15,16,20,36,39				
Q38	A9=B9		38		15,16,18,19,20, 36,39				
Q39	A8=B8		37		15,16,17,18,19, 20,36,38,39				
Q40	A7>B7	22	29	28	2,3,9		300		
Q41	A6>B6		11		2,3,9,29,30				
Q42	A5>B5		14		2,3,9,11,13,29,30				
Q43	A4>B4		12		2,3,9,11,13,14, 29,30,31				
Q44	A3>B3		23		2,3,9,11,12,13, 14,21,29,30,31				
Q45	A2>B2		24		2,3,4,9,11,12,13, 14,21,23,29,30,31				
Q46	A1>B1		5		2,3,4,9,11,12,13, 14,21,23,24,25, 29,30,31				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q47	A0>B0 Propagation delay time (t_{PHL}), C_L = 50 pF	22	6	28	2,3,4,5,9,11,12, 13,14,21,23,24,25, 26,29,30,31	10	300	ns	12, 13
Q48	A7=B7		29	8	2,3,9				
Q49	A6=B6		11		2,3,9,29,30				
Q50	A5=B5		14		2,3,9,11,13,29,30				
Q51	A4=B4		12		2,3,9,11,13,14,29, 30,31				
Q52	A3=B3		23		2,3,9,11,12,13,14, 21,29,30,31				
Q53	A2=B2		24		2,3,4,9,11,12,13, 14,21,23,29,30,31				
Q54	A1=B1		5		2,3,4,9,11,12,13, 14,21,23,24,25,29, 30,31				
Q55	A0=B0		6		2,3,4,5,9,11,12, 13,14,21,23,24,25, 26,29,30,31				
Q56	A7<B7	9,22	29	7	2,3				
Q57	A6<B6	9,22, 29,30	11						
Q58	A5<B5	9,11, 13,22, 29,30	14						
Q59	A4<B4	9,11, 13,14, 22,29, 30,31	12						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q60	A3<B3 Propagation delay time (t_{PHL}), C_L = 50 pF	9,11, 12,13, 14,21, 22,29, 30,31	23	7	2,3	10	300	ns	12, 13
Q61	A2<B2	4,9, 11,12, 13,14, 21,22, 29,30, 31,23	24						
Q62	A1<B1	4,9, 11,12, 13,14, 21,22, 23,24, 25,29, 30,31	5						
Q63	A0<B0	4,5,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	6						
Q64	A7=B7	9,22	29	8					
Q65	A6=B6	9,22, 29,30	11	8					
Q66	A5=B5	9,11, 13,22, 29,30	14	8					

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q67	A4=B4 Propagation delay time (t_{PHL}), C_L = 50 pF	9,11, 13,14, 22,29, 30,31	12	8	2,3	10	300	ns	12, 13
Q68	A3=B3	9,11, 12,13, 14,21, 22,29, 30,31	23						
Q69	A2=B2	4,9, 11,12, 13,14, 21,22, 23,29, 30,31	24						
Q70	A1=B1	4,9, 11,12, 13,14, 21,22, 23,24, 25,29, 30,31	5						
Q71	A0=B0	4,5, 9,11, 12,13, 14,21, 22,23, 24,25, 26,29, 30,31	6						
Q72	A7>B7	22,29	9	28					

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q73	A6>B6 Propagation delay time (t_{PHL}), C_L = 50 pF	9,11, 22,29	30	28	2,3	10	300	ns	12, 13
Q74	A5>B5	9,11, 14,22, 29,30	13						
Q75	A4>B4	9,11, 12,13, 14,22, 29,30	31						
Q76	A3>B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
Q77	A2>B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
Q78	A1>B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						
Q79	A0>B0	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	26						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q80	A7=B7 Propagation delay time (t_{PHL}), C_L = 50 pF	22,29	9	8	2,3	10	300	ns	12, 13
Q81	A6=B6	9,11, 22,29	30						
Q82	A5=B5	9,11, 14,22, 29,30	13						
Q83	A4=B4	9,11, 12,13, 14,22, 29,30	31						
Q84	A3=B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
Q85	A2=B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
Q86	A1=B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q87	A0=B0 Propagation delay time (t_{PHL}), C_L = 50 pF	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	26	8	2,3	10	300	ns	12, 13
Q88	A7<B7	22	9	7	2,3,29				
Q89	A6<B6		30		2,3,9,11,29				
Q90	A5<B5		13		2,3,9,11,14,29,30				
Q91	A4<B4		31		2,3,9,11,12,13, 14,29,30				
Q92	A3<B3		21		2,3,9,11,12,13, 14,23,29,30,31				
Q93	A2<B2		4		2,3,9,11,12,13, 14,21,23,24,29, 30,31				
Q94	A1<B1		25		2,3,4,5,9,11,12, 13,14,21,23,24, 29,30,31				
Q95	A0<B0		26		2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,29,30,31				
Q96	A7=B7		9	8	2,3,29				
Q97	A6=B6		30	8	2,3,9,11,29				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q98	A5=B5 Propagation delay time (t_{PHL}), C_L = 50 pF	22	13	8	2,3,9,11,14,29,30	10	300	ns	12, 13
Q99	A4=B4		31		2,3,9,11,12,13,14, 29,30				
Q100	A3=B3		21		2,3,9,11,12,13,14, 23,29,30,31				
Q101	A2=B2		4		2,3,9,11,12,13,14, 21,23,24,29,30,31				
Q102	A1=B1		25		2,3,4,5,9,11,12, 13,14,21,23,24, 29,30,31				
Q103	A0=B0		26		2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 29,30,31				
R1	A<B IN 1 to A<B OUT 1 Propagation delay time (t_{PLH}), C_L = 150 pF	Open	2	7	3,4,5,6,9,11,12, 13,14,21,22,23,24, 25,26,29,30,31		320		
R2	A=B IN 1 to A=B OUT 1		22	8	2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 26,29,30,31		320		
R3	A>B IN 1 to A>B OUT 1		3	28	2,4,5,6,9,11,12, 13,14,21,22,23,24, 25,26,29,30,31		320		
R4	A=B IN 1 to A=B MON		22	27	2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 26,29,30,31		170		
R5	A<B IN 2 to A<B OUT 2		36	32	15,16,17,18,19,20, 35,37,38,39		170		

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TABLE IX. Transition time (t_{THL} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R6	A=B IN 2 to A=B OUT 2 Propagation delay time (t_{PLH}), C_L = 150 pF	Open	35	33	15,16,17,18,19,20, 36,37,38,39	10	170	ns	12, 13
R7	A>B IN 2 to A>B OUT 2	Open	15	34	16,17,18,19,20,35, 36,37,38,39				
R8	A11>B11	35	20		15,16,36				
R9	A10>B10		39		15,16,19,20,36				
R10	A9>B9		18		15,16,19,20,36, 38,39				
R11	A8>B8		17		15,16,18,19,20, 36,37,38,39				
R12	A11=B11		20	33	15,16,36				
R13	A10=B10		39		15,16,19,20,36				
R14	A9=B9		18		15,16,19,20,36, 38,39				
R15	A8=B8		17		15,16,18,19,20,36, 37,38,39				
R16	A11<B11	16,35	20	32	15,36				
R17	A10<B10	16,19, 20,35	39	32	15,36				
R18	A9<B9	16,19, 20,35, 38,39	18	32	15,16				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R19	A8<B8 Propagation delay time (t_{PLH}), C_L = 150 pF	16,18, 19,20, 35,37, 38,39	17	32	15,36	10	170	ns	12, 13
R20	A11=B11	16,35	20	33					
R21	A10=B10	16,19, 20,35	39						
R22	A9=B9	16,19, 20,35, 38,39	18						
R23	A8=B8	16,18, 19,20, 35,37, 38,39	17						
R24	A11>B11	20,35	16	34					
R25	A10>B10	16,20, 39,35	19						
R26	A9>B9	16,18, 19,20, 35,39	38						
R27	A8>B8	16,17, 18,19, 20,35, 38,39	37						
R28	A11=B11	20,35	16	33					
R29	A10=B10	16,20, 35,39	19	33					

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R30	A9=B9 Propagation delay time (t_{PLH}), C_L = 150 pF	16,18, 19,20, 35,39	38	33	15,36	10	170	ns	12, 13
R31	A8=B8	16,17, 18,19, 20,35, 38,39	37	33	15,16				
R32	A11<B11	35	16	32	15,20,36				
R33	A10<B10		19		15,16,20,36,39				
R34	A9<B9		38		15,16,18,19,20,36, 39				
R35	A8<B8		37		15,16,17,18,19,20, 36,38,39				
R36	A11=B11		16	33	15,20,36				
R37	A10=B10		19		15,16,20,36,39				
R38	A9=B9		38		15,16,18,19,20,36, 39				
R39	A8=B8		37		15,16,17,18,19,20, 36,38,39				
R40	A7>B7	22	29	28	2,3,9		320		
R41	A6>B6		11		2,3,9,29,30				
R42	A5>B5		14		2,3,9,11,13,29,30				
R43	A4>B4		12		2,3,9,11,13,14, 29,30,31				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R44	A3>B3 Propagation delay time (t_{PLH}), C_L = 150 pF	22	23	28	2,3,9,11,12,13,14, 21,29,30,31	10	320	ns	12, 13
R45	A2>B2		24		2,3,4,9,11,12,13, 14,21,23,29,30,31				
R46	A1>B1		5		2,3,4,9,11,12,13, 14,21,23,24,25,29, 30,31				
R47	A0>B0		6		2,3,4,5,9,11,12, 13,14,21,23,24,25, 26,29,30,31				
R48	A7=B7		29	8	2,3,9				
R49	A6=B6		11		2,3,9,29,30				
R50	A5=B5		14		2,3,9,11,13,29,30				
R51	A4=B4		12		2,3,9,11,13,14, 29,30,31				
R52	A3=B3		23		2,3,9,11,12,13, 14,21,29,30,31				
R53	A2=B2		24		2,3,4,9,11,12,13, 14,21,23,29,30,31				
R54	A1=B1		5		2,3,4,9,11,12,13, 14,21,23,24,25, 29,30,31				
R55	A0=B0		6		2,3,4,5,9,11,12, 13,14,21,23,24,25, 26,29,30,31				
R56	A7<B7	9,22	29	7	2,3				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R57	A6<B6 Propagation delay time (t_{PLH}), C_L = 150 pF	9,22, 29,30	11	7	2,3	10	320	ns	12, 13
R58	A5<B5	9,11, 13,22, 29,30	14						
R59	A4<B4	9,11, 13,14, 22,29, 30,31	12						
R60	A3<B3	9,11, 12,13, 14,21, 22,29, 30,31	23						
R61	A2<B2	4,9,11, 12,13, 14,21, 22,29, 30,31, 23	24						
R62	A1<B1	4,9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	5						
R63	A0<B0	4,5,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	6						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R64	A7=B7 Propagation delay time (t_{PLH}), C_L = 150 pF	9,22	29	8	2,3	10	320	ns	12, 13
R65	A6=B6	9,22, 29,30	11						
R66	A5=B5	9,11, 13,22, 29,30	14						
R67	A4=B4	9,11, 13,14, 22,29, 30,31	12						
R68	A3=B3	9,11, 12,13, 14,21, 22,29, 30,31	23						
R69	A2=B2	4,9,11, 12,13, 14,21, 22,23, 29,30, 31	24						
R70	A1=B1	4,9, 11,12, 13,14, 21,22, 23,24, 25,29, 30,31	5						
R71	A0=B0	4,5,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	6						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R72	A7>B7 Propagation delay time (t_{PLH}), C_L = 150 pF	22,29	9	28	2,3	10	320	ns	12, 13
R73	A6>B6	9,11, 22,29	30						
R74	A5>B5	9,11, 14,22, 29,30	13						
R75	A4>B4	9,11, 12,13, 14,22, 29,30	31						
R76	A3>B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
R77	A2>B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
R78	A1>B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						
R79	A0>B0	4,5,6,9 11,12, 13,14, 21,22, 23,24, 25,29, 30,31	26						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R80	A7=B7 Propagation delay time (t_{PLH}), C_L = 150 pF	22,29	9	8	2,3	10	320	ns	12, 13
R81	A6=B6	9,11, 22,29	30						
R82	A5=B5	9,11, 14,22, 29,30	13						
R83	A4=B4	9,11, 12,13, 14,22, 29,30	31						
R84	A3=B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
R85	A2=B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
R86	A1=B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						
R87	A0=B0	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	26						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R88	A7<B7 Propagation delay time (t_{PLH}), C_L = 150 pF	22	9	7	2,3,29	10	320	ns	12, 13
R89	A6<B6		30		2,3,9,11,29				
R90	A5<B5		13		2,3,9,11,14,29,30				
R91	A4<B4		31		2,3,9,11,12,13,14 29,30				
R92	A3<B3		21		2,3,9,11,12,13,14 23,29,30,31				
R93	A2<B2		4		2,3,9,11,12,13,14 21,23,24,29,30,31				
R94	A1<B1		25		2,3,4,5,9,11,12, 13,14,21,23,24,29 30,31				
R95	A0<B0		26	7	2,3,4,5,6,9,11,12 13,14,21,23,24,25 29,30,31				
R96	A7=B7		9	8	2,3,29				
R97	A6=B6		30		2,3,9,11,29				
R98	A5=B5		13		2,3,9,11,14,29,30				
R99	A4=B4		31		2,3,9,11,12,13,14 29,30				
R100	A3=B3		21		2,3,9,11,12,13,14 23,29,30,31				
R101	A2=B2		4		2,3,9,11,12,13,14 21,23,24,29,30,31				
R102	A1=B1		25		2,3,4,5,9,11,12, 13,14,21,23,24,29 30,31				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R103	A0=B0 Propagation delay time (t_{PLH}), C_L = 150 pF	22	26	8	2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 29,30,31	10	320	ns	12, 13
S1	A<B IN 1 to A<B OUT 1 Propagation delay time (t_{PHL}), C_L = 150 pF	Open	2	7	3,4,5,6,9,11,12, 13,14,21,22,23,24, 25,26,29,30,31		220		
S2	A=B IN 1 to A=B OUT 1		22	8	2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 26,29,30,31		220		
S3	A>B IN 1 to A>B OUT 1		3	28	2,4,5,6,9,11,12, 13,14,21,22,23,24, 25,26,29,30,31		220		
S4	A=B IN 1 to A=B MON		22	27	2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 26,29,30,31		120		
S5	A<B IN 2 to A<B OUT 2		36	32	15,16,17,18,19,20, 35,37,38,39				
S6	A=B IN 2 to A=B OUT 2		35	33	15,16,17,18,19,20, 36,37,38,39				
S7	A>B IN 2 to A>B OUT 2		15	34	16,17,18,19,20,35, 36,37,38,39				
S8	A11>B11	35	20	34	15,16,36		170		
S9	A10>B10	35	39	34	15,16,19,20,36		170		

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S10	A9>B9 Propagation delay time (t_{PHL}), C_L = 150 pF	35	18	34	15,16,19,20,36,38,39	10	170	ns	12, 13
S11	A8>B8		17	34	15,16,18,19,20,36,37,38,39				
S12	A11=B11		20	33	15,16,36				
S13	A10=B10		39		15,16,19,20,36				
S14	A9=B9		18		15,16,19,20,36,38,39				
S15	A8=B8		17		15,16,18,19,20,36,37,38,39				
S16	A11<B11	16,35	20	32	15,36				
S17	A10<B10	16,19,20,35	39						
S18	A9<B9	16,19,20,35,38,39	18						
S19	A8<B8	16,18,19,20,35,37,38,39	17						
S20	A11=B11	16,35	20	33					
S21	A10=B10	16,19,20,35	39	33					
S22	A9=B9	16,19,20,35,38,39	18	33					

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S23	A8=B8 Propgation delay time (t_{PHL}), C_L = 150 pF	16,18, 19,20, 35,37, 38,39	17	33	15,36	10	170	ns	12, 13
S24	A11>B11	20,35	16	34					
S25	A10>B10	16,20, 39,35	19						
S26	A9>B9	16,18, 19,20, 35,39	38						
S27	A8>B8	16,17, 18,19, 20,35, 38,39	37						
S28	A11=B11	20,35	16	33					
S29	A10=B10	16,20, 35,39	19						
S30	A9=B9	16,18, 19,20, 35,39	38						
S31	A8=B8	16,17, 18,19, 20,35, 38,39	37						
S32	A11<B11	35	16	32	15,20,36				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S33	A10<B10 Propagation delay time (t_{PHL}), C_L = 150 pF	35	19	32	15,16,20,36,39	10	170	ns	12, 13
S34	A9<B9		38	32	15,16,18,19,20, 36,39				
S35	A8<B8		37	32	15,16,17,18,19, 20,36,38,39				
S36	A11=B11		16	33	15,20,36				
S37	A10=B10		19		15,16,20,36,39				
S38	A9=B9		38		15,16,18,19,20,36, 39				
S39	A8=B8		37		15,16,17,18,19,20, 36,38,39				
S40	A7>B7	22	29	28	2,3,9		320		
S41	A6>B6		11		2,3,9,29,30				
S42	A5>B5		14		2,3,9,11,13,29,30				
S43	A4>B4		12		2,3,9,11,13,14, 29,30,31				
S44	A3>B3		23		2,3,9,11,12,13,14, 21,29,30,31				
S45	A2>B2		24		2,3,4,9,11,12,13, 14,21,23,29,30,31				
S46	A1>B1		5		2,3,4,9,11,12,13, 14,21,23,24,25,29, 30,31				

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S47	A0>B0 Propgation delay time (t_{PHL}), C_L = 150 pF	22	6	28	2,3,4,5,9,11,12, 13,14,21,23,24, 25,26,29,30,31	10	320	ns	12, 13
S48	A7=B7		29	8	2,3,9				
S49	A6=B6		11		2,3,9,29,30				
S50	A5=B5		14		2,3,9,11,13,29,30				
S51	A4=B4		12		2,3,9,11,13,14,29, 30,31				
S52	A3=B3		23		2,3,9,11,12,13,14, 21,29,30,31				
S53	A2=B2		24		2,3,4,9,11,12,13, 14,21,23,29,30,31				
S54	A1=B1		5		2,3,4,9,11,12,13, 14,21,23,24,25,29, 30,31				
S55	A0=B0		6		2,3,4,5,9,11,12, 13,14,21,23,24,25, 26,29,30,31				
S56	A7<B7	9,22	29	7	2,3				
S57	A6<B6	9,22, 29,30	11						
S58	A5<B5	9,11, 13,22, 29,30	14						
S59	A4<B4	9,11, 13,14, 22,29, 30,31	12						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S60	A3<B3 Propagation delay time (t_{PHL}), C_L = 150 pF	9,11, 12,13, 14,21, 22,29, 30,31	23	7	2,3	10	320	ns	12, 13
S61	A2<B2	4,9,11, 12,13, 14,21, 22,29, 30,31, 23	24						
S62	A1<B1	4,9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	5						
S63	A0<B0	4,5,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	6						
S64	A7=B7	9,22	29	8					
S65	A6=B6	9,22, 29,30	11						
S66	A5=B5	9,11, 13,22, 29,30	14	8					

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S67	A4=B4 Propagation delay time (t_{PHL}), C_L = 150 pF	9,11, 13,14, 22,29, 30,31	12	8	2,3	10	320	ns	12, 13
S68	A3=B3	9,11, 12,13, 14,21, 22,29, 30,31	23						
S69	A2=B2	4,9,11, 12,13, 14,21, 22,23, 29,30, 31	24						
S70	A1=B1	4,9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	5						
S71	A0=B0	4,5,9, 11,12, 13,14, 21,22, 23,24, 25,26, 29,30, 31	6						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S72	A7>B7 Propagation delay time (t_{PHL}), C_L = 150 pF	22,29	9	28	2,3	10	320	ns	12, 13
S73	A6>B6	9,11, 22,29	30						
S74	A5>B5	9,11, 14,22, 29,30	13						
S75	A4>B4	9,11, 12,13, 14,22, 29,30	31						
S76	A3>B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
S77	A2>B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
S78	A1>B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						
S79	A0>B0	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	26						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

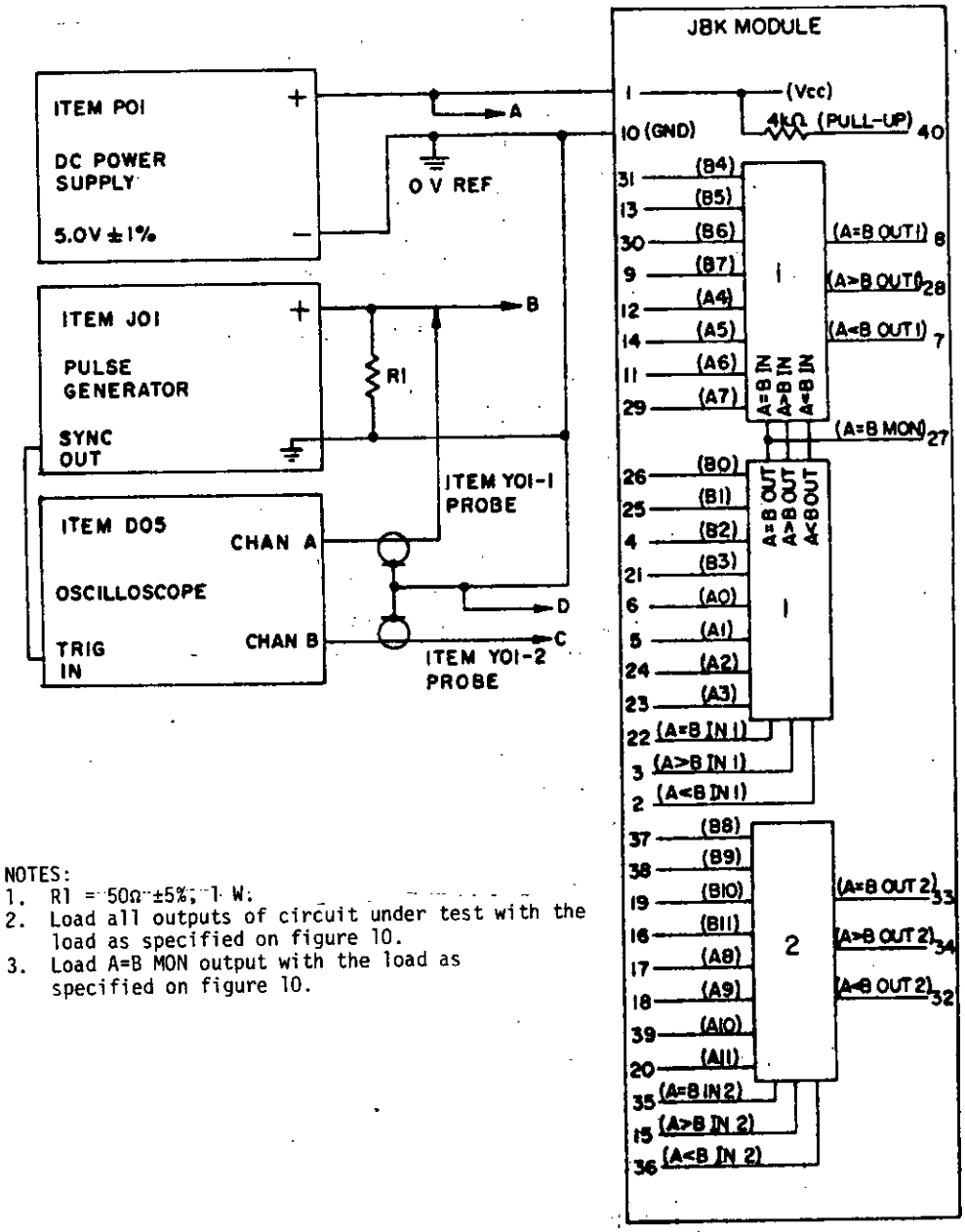
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S80	A7=B7 Propagation delay time (t_{PHL}), C_L = 150 pF	22,29	9	8	2,3	10	320	ns	12, 13
S81	A6=B6	9,11, 22,29	30						
S82	A5=B5	9,11, 14,22, 29,30	13						
S83	A4=B4	9,11, 12,13, 14,22, 29,30	31						
S84	A3=B3	9,11, 12,13, 14,22, 23,29, 30,31	21						
S85	A2=B2	9,11, 12,13, 14,21, 22,23, 24,29, 30,31	4						
S86	A1=B1	4,5,9, 11,12, 13,14, 21,22, 23,24, 29,30, 31	25						
S87	A0=B0	4,5,6, 9,11, 12,13, 14,21, 22,23, 24,25, 29,30, 31	26						

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TABLE IX. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S88	A7<B7 Propagation delay time (t_{PHL}), C_L = 150 pF	22	9	7	2,3,29	10	320	ns	12, 13
S89	A6<B6		30		2,3,9,11,29				
S90	A5<B5		13		2,3,9,11,14,29,30				
S91	A4<B4		31		2,3,9,11,12,13,14, 29,30				
S92	A3<B3		21		2,3,9,11,12,13,14, 23,29,30,31				
S93	A2<B2		4		2,3,9,11,12,13,14, 21,23,24,29,30,31				
S94	A1<B1		25		2,3,4,5,9,11,12, 13,14,21,23,24,29, 30,31				
S95	A0<B0		26		2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 29,30,31				
S96	A7=B7		9	8	2,3,29				
S97	A6=B6		30		2,3,9,11,29				
S98	A5=B5		13		2,3,9,11,14,29,30				
S99	A4=B4		31		2,3,9,11,12,13,14, 29,30				
S100	A3=B3		21		2,3,9,11,12,13,14, 23,29,30,31				
S101	A2=B2		4		2,3,9,11,12,13,14, 21,23,24,29,30,31				
S102	A1=B1		25		2,3,4,5,9,11,12, 13,14,21,23,24,29, 30,31				
S103	A0=B0		26		2,3,4,5,6,9,11,12, 13,14,21,23,24,25, 29,30,31				

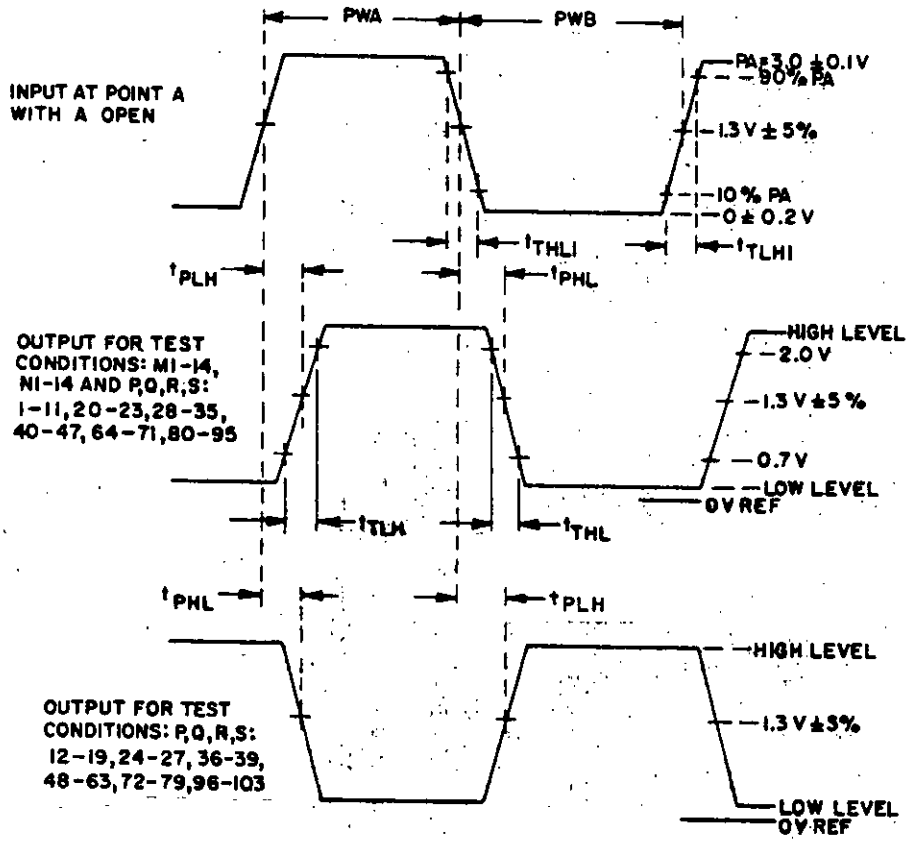
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load all outputs of circuit under test with the load as specified on figure 10.
3. Load A=B MON output with the load as specified on figure 10.

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table IX.

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NOTES:
1. Input pulse parameters:

   (a) PWA = 400 ±20 ns.
   (b) PWB = 600 ±60 ns.
   (c) $t_{TLH_1} = t_{THL_1} = 25 \pm 2$ ns.

2. Before making measurement, the output levels shall be verified to be 0.3 V maximum for a low level and 2.4 V minimum for a high level.
3. All voltage levels are referenced to ground.

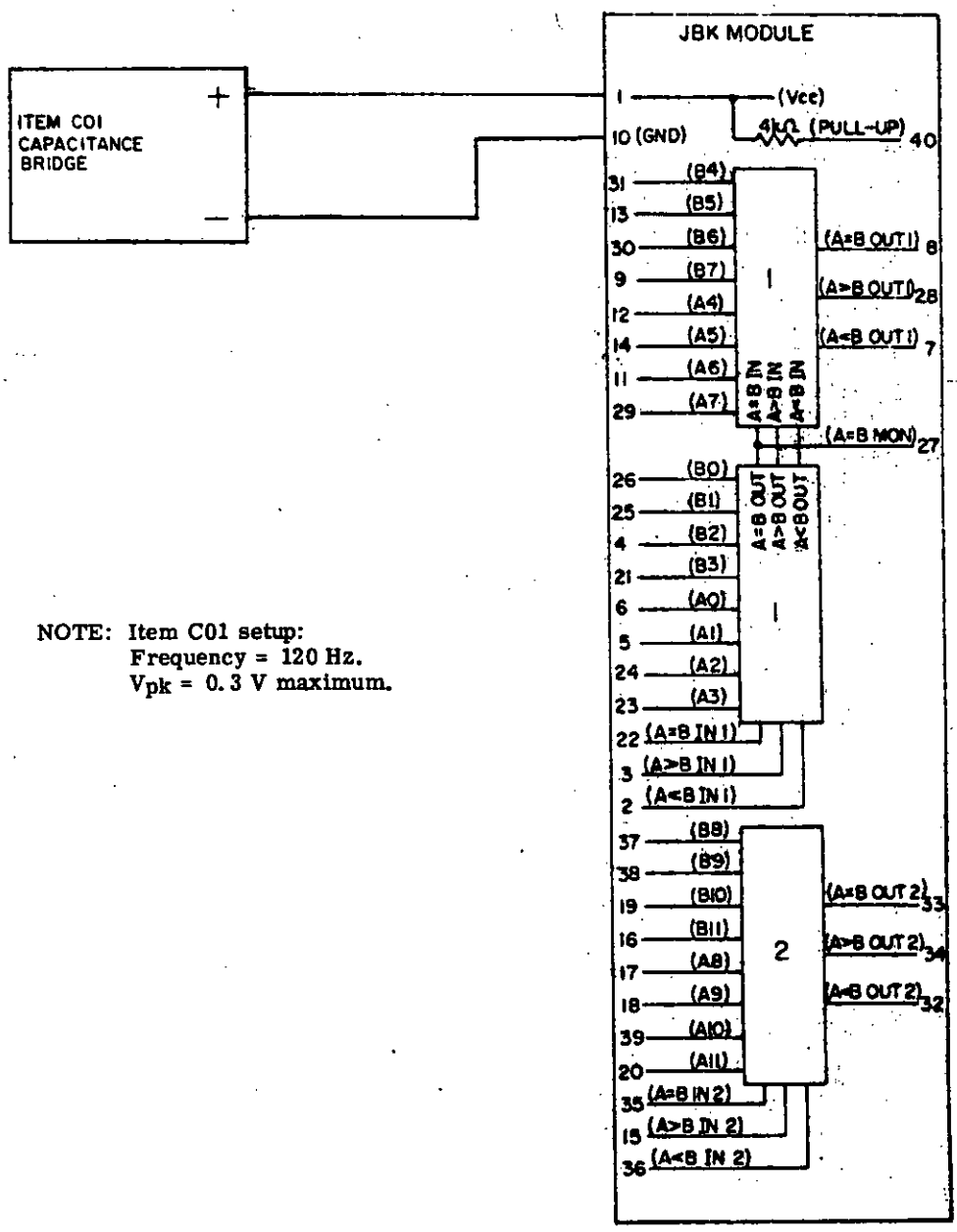
FIGURE 13. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table IX.

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TABLE X. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply filter capacitance	1.6	3.0	µF	14

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NOTE: Item C01 setup:
Frequency = 120 Hz.
Vpk = 0.3 V maximum.

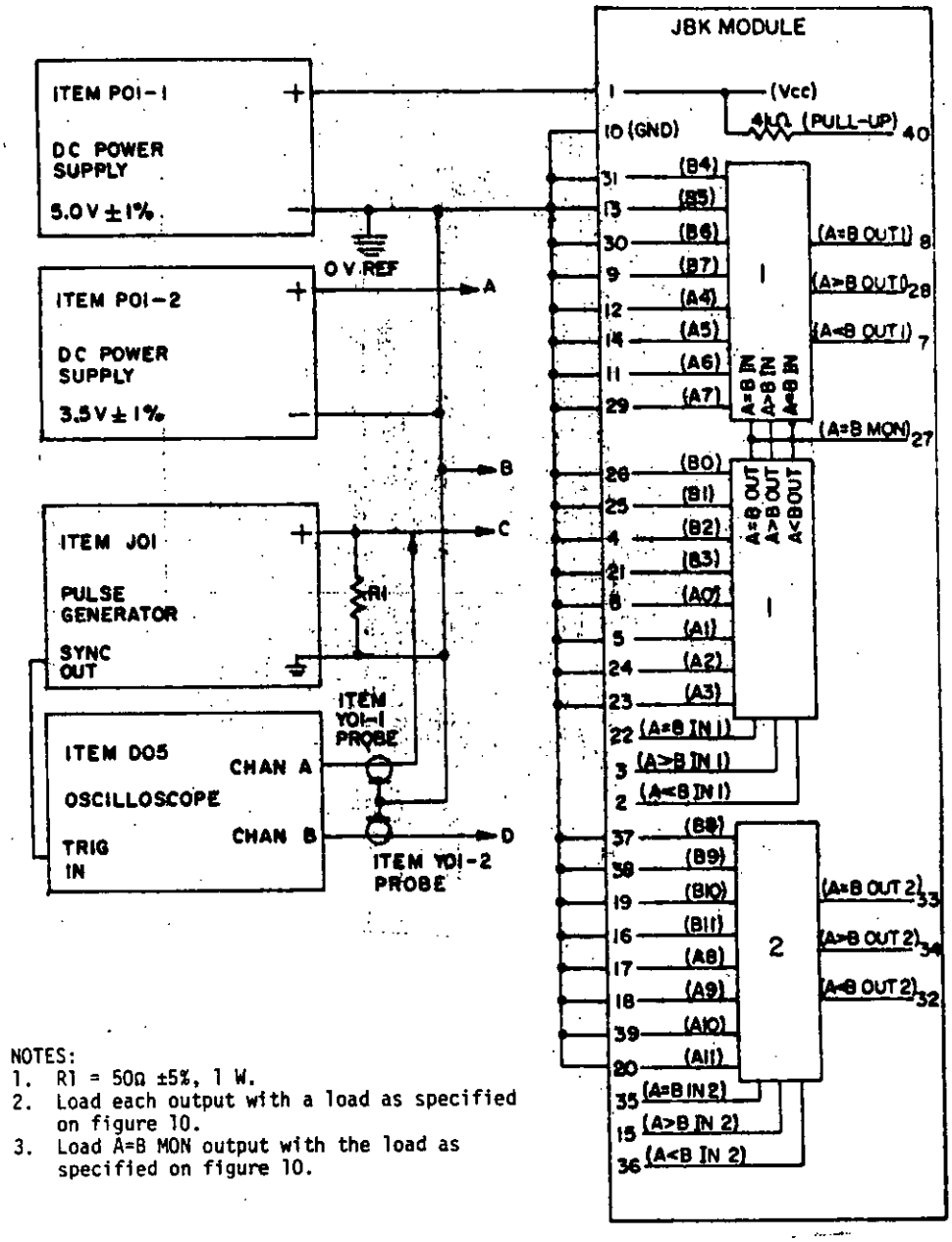
FIGURE 14. Power supply filter capacitance test circuit for table X.

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TABLE XI. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	A>B OUT 1 Output crosstalk, high level	3	Open	2,22, 15,35, 36	28	0	400	mV	15, 16
U2	A<B OUT 1	2		3,22, 15,35, 36	7				
U3	A=B OUT 1	22		2,3,15, 35,36	8				
U4	A=B MON	22		2,3,15, 35,36	27				
U5	A>B OUT 2	15		2,3,22, 35,36	34				
U6	A<B OUT 2	36		2,3,22, 15,35	32				
U7	A=B OUT 2	35		2,3,22, 15,36	33				
V1	A>B OUT 1 Output crosstalk, low level	Open	3	2,22, 15,35, 36	28				
V2	A<B OUT 1		2	3,22,15, 35,36	7				
V3	A=B OUT 1		22	2,3,15, 35,36	8				
V4	A=B MON		22	2,3,15, 35,36	27				
V5	A>B OUT 2		15	2,3,22, 35,36	34				
V6	A<B OUT 2		36	2,3,22, 15,35	32				
V7	A=B OUT 2		35	2,3,22, 15,36	33				

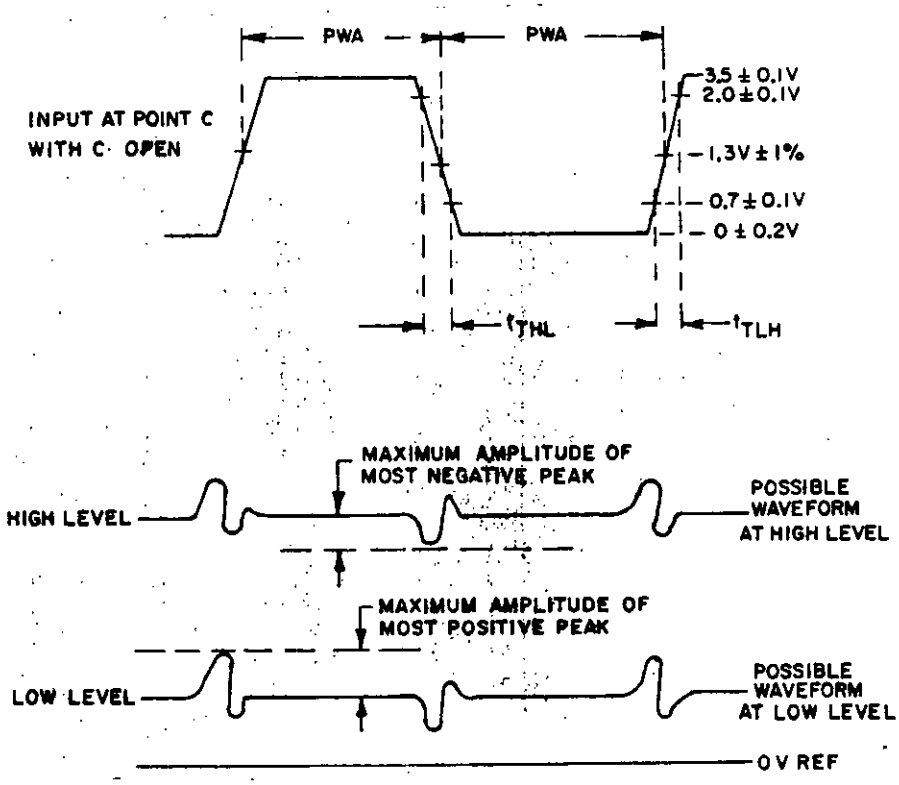
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 10.
3. Load A=B MON output with the load as specified on figure 10.

FIGURE 15. Output crosstalk (high-level and low-level) test circuit for table XI.

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NOTE: Input pulse parameters:

(a) PWA = 500 ±50 ns.
(b) $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns. (verified at each input pin).

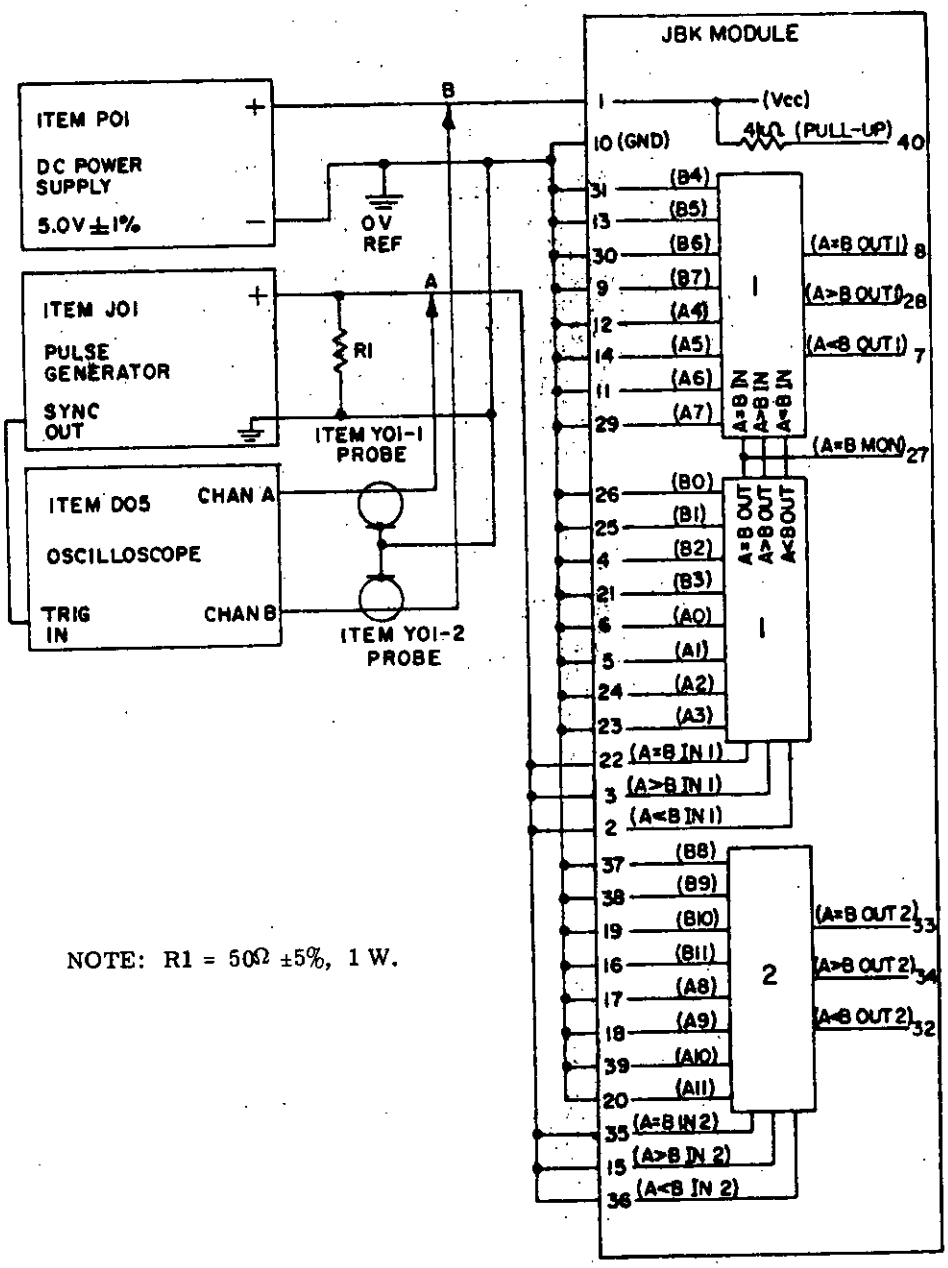
FIGURE 16. Output crosstalk (high-level and low-level) test circuit waveforms for table XI.

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TABLE XII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
W	Power supply transient amplitude	0	400	mV	17, 18

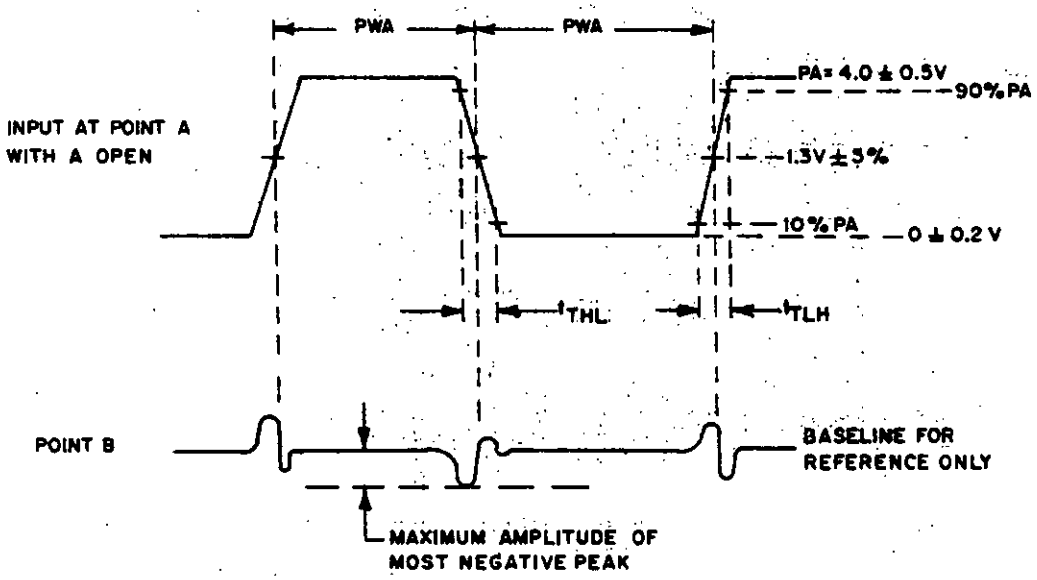
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NOTE: R1 = 50Ω ±5%, 1 W.

FIGURE 17. Power supply transient amplitude test circuit for table XII.

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NOTE: Input pulse parameters:

(a) PWA = 500 ±50 ns.
(b) $t_{TLH} = t_{THL} = 25 \pm 2$ ns.

FIGURE 18. Power supply transient amplitude test circuit waveforms for table XII.

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TABLE XIII. Input capacitance test.

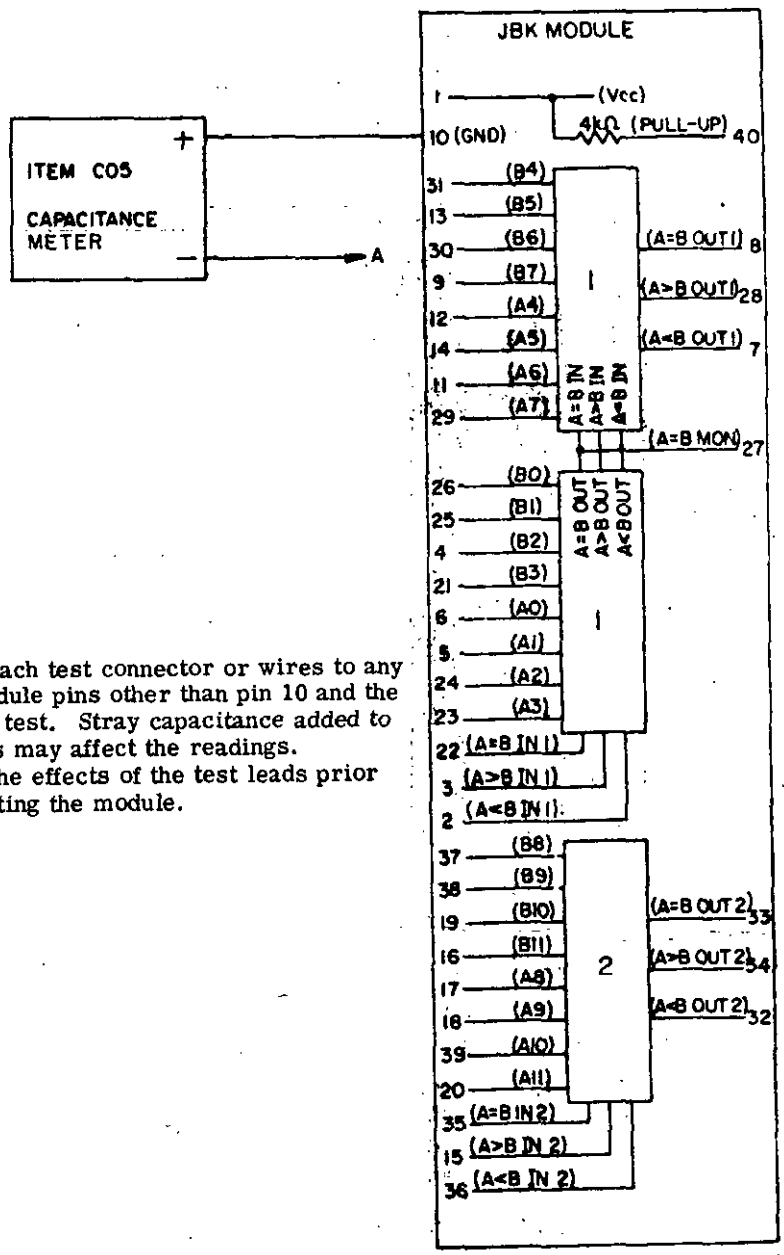
Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
X1	A>B IN 1 Input capacitance	3	2	15	pF	19
X2	A<B IN 1	2				
X3	A>B IN 2	15				
X4	A<B IN 2	36				
X5	A=B IN 1	22				
X6	A=B IN 2	35				
X7	A0	6		30		
X8	A1	5				
X9	A2	24				
X10	A3	23				
X11	A4	12				
X12	A5	14				
X13	A6	11				
X14	A7	29				
X15	A8	17				
X16	A9	18				
X17	A10	39				
X18	A11	20				
X19	B0	26				
X20	B1	25				
X21	B2	4				
X22	B3	21				
X23	B4	31				
X24	B5	13				
X25	B6	30				

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TABLE XIII. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
X26	B7 Input capacitance	9	2	30	pF	19
X27	B8	37	↓	↓	↓	↓
X28	B9	38				
X29	B10	19				
X30	B11	16				

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 19. Input capacitance test circuit for table XIII.

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